

Order Flow Toxicity and Informed Trading Around Known Market Manipulation Events: Evidence from Interest Rate Futures

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Abstract

The objective of this paper is an ex-post review of the effectiveness of PIN and VPIN in determining changes in the information structure and order flow of a futures market around documented episodes of recorded manipulation of the reference rate, from the various publicly available regulatory reports. In keeping with previous studies on interest rate derivatives, this analysis finds that the average PIN is far higher for futures than for the equity market at or above 60%. Furthermore, I find a very strong connection between PIN, VPIN and time to maturity of the contract that is not fully explained by the time variation in activity in the market. However, an event study using both a new bootstrap approach and asymptotic standard error on the VPIN and PIN respectively around documented LIBOR manipulation cases has mixed results. For certain events I see a substantial change in the average detected levels of PIN and VPIN, however a cross sectional analysis of all reported cases up to mid-2015, indicates no significant change in the PIN and VPIN for the contracts in this Eurodollar sample.

Keywords: Microstructure, Informed Trading, PIN, VPIN, LIBOR, Eurodollar, Market Manipulation

1. Introduction

There are currently two key debates ongoing in the financial community related to issues around market manipulation. First, in the policy and practitioner community, there is evidence of manipulation of key benchmarks by financial institutions, and there is some disagreement as to whether this occurred primarily to improve their trading opportunities or to reduce the perception of balance sheet weakness. Second, is an academic debate as to whether the current high profile measures of the probability-of-informed trading (PIN)

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and ‘market-toxicity’ (measured by a volume adjusted PIN denoted VPIN) actually provide substantive measurement of the phenomena in question.

This study takes a comprehensive dataset from Chicago Mercantile Exchange (CME) tape data, which covers every inside quote and trade from 1996 to 2015 for the 40 LIBOR referenced quarterly dated Eurodollar futures contracts. First, I apply different flavours of the PIN and VPIN metrics over a variety of estimation windows. A variety of tests is constructed to see if the pattern of the PIN and VPIN exhibit structural changes around documented cases of manipulation of the LIBOR reference rate. Finally, I compare this to systematic fluctuations in these measures relative to the futures term structure.

Unsurprisingly, given the scale of the task, this study finds very mixed results. Both PIN and VPIN vary systematically and in a statistically significant pattern in respect to the term structure of the futures contracts. PIN varies in a v-shaped pattern, with long (2000 to 3500 days) and short maturity (0 to 500 days) contracts having significantly higher PIN and VPIN measurements than intermediate contracts (which are actually the most heavily traded). VPIN measurements of informed trading are substantially lower, over the entire gamut of calculation measures. However, when I move to documented cases of market manipulation in the reference rate, the results are ambiguous. There are definitive cases when the PIN (and to an extent certain flavours of the VPIN) shift systematically around a relevant documented LIBOR manipulation event. However, when I build cross sectional averages across events, I find no significant evidence of systematic shifts in either the PIN or VPIN metric. It should be noted that whilst I have included every documented case of manipulation directly linked to the relevant reference rates, my list is necessarily incomplete as the regulatory actions have tended to focus on sample charges to the firms involved, rather than documenting every occurrence and its motivation.

The contribution of this chapter is threefold. First, this is the only paper to provide a comprehensive analysis of informed trading in Eurodollar futures from 1996 to 2015. Second,

to implement my analysis I have introduced several innovations in the estimation process, including an exact asymptotic representation of the measurement error of the PIN and a new bootstrap-in-bootstrap method for estimation of the VPIN. Finally, my methodological approach to decomposing informed trading by both event and the term-structure is a new contribution to this literature and my algorithms are available for other researchers to implement such studies in different market settings.

The remainder of this paper is organized as follows. First, §.2 outlines the debate on informed trading, in the academic literature in regard to the efficacy of the PIN and more recent VPIN algorithms. To this end, §.3 I look more deeply at the adaptation of the Glosten and Milgrom (1985) model of informed trading and demonstrate how a version of the classic version of the PIN could be used to empirically fit this model, in the spirit of Glosten and Milgrom (1985) and Easley et al. (1996). In this section, I document the various empirical strategies to estimate the PIN and VPIN including several new features specific to this study relating to the distribution of these critical statistics. §.4 provides a detailed of Eurodollar futures and the descriptive of the term-structure of PIN and PIN's parameters. Subsequently §.5 presents the analysis of my dataset. For each contract maturity, the PIN and VPIN is estimated and I construct experiments by aggregation across the terms structure, in addition to using the LIBOR manipulation events discussed previously. Finally, in §.6 presents the summaries of the key findings and some brief concluding comments and directions for future research.

2. The VPIN Debate and Further Background

The PIN model was first introduced in Easley and O'Hara (1992) and Easley et al. (1996) and estimated over relatively high frequency data for US equities. Subsequent work focused on the PIN as an asset pricing factor or related the analysis directly to corporate

governance. The theoretical model underlying the PIN evolved from the information based trading literature that started with Grossman and Stiglitz (1980) and Hellwig (1980) then continued with Admati (1985), Hasbrouck (1991) and Kyle (1985). However, the most direct theoretical antecedent to the PIN model is Glosten and Milgrom (1985) and this is the starting point for interpreting the impact of LIBOR manipulation on the Eurodollar futures market. In general the theoretical models of informed trading have traveled down the standard roots, Glosten and Milgrom (1985) and Kyle (1985) have used Bayesian games to analyse simple information flows in single assets, whereas Hellwig (1980), Admati (1985) and Admati and Pfleiderer (1988) have utilized a rational expectations framework for single and multiple assets.

However, regardless of the solution device applied, either rational expectations or a Bayes-Nash equilibrium, the general set-up is very similar. First, certain traders are provided with a forward looking information endowment (usually a noisy forecast of direction) and then there is a trading game against another group of traders who either lack this information or whose order flow is constrained. For instance, in the case of Eurodollar futures, we know that floating legs of IRSs are offset at or around international money market (IMM) dates, hence one particular institution may not have any choice in the timing and net quantity of futures demanded whether long or short. Hence a trader with these constraints is a net supplier of random order-flow to the market regardless of how the term structure is updated by the new fixings.

One of the unique selling points of the PIN model is the ease of implementation. The Glosten and Milgrom (1985) model is easily to re-cast in terms of net order flow and hence one merely needs to sign each trade as a buy or a sell to determine the net order-flow imbalance. The PIN is then computed via maximum-likelihood estimation. However, signing trades as a buy or a sell can be a difficult process. The standard Lee and Ready (1991) algorithm matches trades based on the distance between the transacted price and the nearest quotes.

Several prior studies have identified difficulties in estimating the PIN attributing most of the problems to trade-classification, see Ke (2014); Yan and Zhang (2012); Boehmer et al. (2007). It is evident that when switching to very high frequency data (such as that generated by futures trading) classifying trades becomes ever more problematic.

As a solution to the PIN classification problem, Easley et al. (2012) proposed a volume based approach known as VPIN, and explicitly introduced the notion of order-flow-toxicity. Order-flow-toxicity is defined as a systematic imbalance in the buy and sell side volume of the order-book that leads to substantial adjustments in spread. This is in opposition to the normal state of affairs where the liquidity would remain relatively constant. However, the proposed application to the May 2010 ‘flash crash’ has proved somewhat controversial. In many respects the ease of implementing the VPIN and the unambiguous nature of its measurement of market data, forms part of the issue. Andersen and Bondarenko (2013) re-computed the VPIN metric for S&P 500 E-mini futures and refuted the applicability of the method to forecast extreme liquidity events. Andersen and Bondarenko (2013) concluded that the VPIN was highly sensitive to the assumptions placed on discretizing the volume of orders, reducing its efficacy as a regulatory instrument for predicting order-flow-toxicity exceed events.

In a series of follow up papers Andersen and Bondarenko (2014c), Easley et al. (2014) and again Andersen and Bondarenko (2014b) debated the relative merits of the VPIN approach. In another follow-up, Andersen and Bondarenko (2014a) proposed an alternative approach to specifically capture order-flow-toxicity. My experience with estimating VPIN falls somewhere between Andersen and Bondarenko (2014c) and Easley et al. (2014). VPIN is indeed sensitive to the choice on how to discretize volume. However, I do find the change in VPIN, and I denote this ΔVPIN which is consistent across a variety of choices. I do not find VPIN and the original PIN to be correlated under my trade-classification approach. Furthermore, I do not find that the VPIN lies within a 95% confidence bound computed

from the maximum likelihood estimates for the underlying parameters. This indicates that for the purposes of identification of informed trading in Eurodollar futures markets, (which uses the same platform as the S&P 500 E-mini), the PIN and VPIN are, unfortunately, incongruent across a wide variety of specifications (including a bootstrapped version).

3. The model

3.1. The Market Algorithm

This section explains how the LIBOR manipulation affects the Eurodollar trading by deriving the fundamentals of sequential trading with superior information proposed by Glosten and Milgrom (1985). This model focuses on stylized leadership market in which a Eurodollar price will reach at time T with value $S_T = \{\underline{S}, \bar{S}\}$, where $\underline{S} \leq \bar{S}$. Uninformed traders (ε) know the Eurodollar price will be $\underline{S} \leq S_t \leq \bar{S}$, however, they do not know either $S_t = \underline{S}$ or $S_t = \bar{S}$ will occur, so these traders are equally likely to buy or sell. Unlikely uninformed traders, informed traders (μ) are said to be informed of knowing the terminal value of the asset S_T . This sequential trade can be explained in terms of probability ($\mathbb{P}$) which is the one that makes most sense. With theoretical explanations incorporated with ED trading, on each trading period traders arrive on the market sequentially in which knowing the value of ED futures equals S_T , suppose that $\mathbb{P}(S_T = \underline{S}) = \delta$ and $\mathbb{P}(S_T = \bar{S}) = 1 - \delta$, which trade occurs at time $t \in [0, T)$. The probability pay off for each possible scenario of S_T is, if the ED expected price is equal $S_T = \underline{S} \leq S_t$ informed traders will wait on the sell side with probability $\delta\mu$, as uninformed Traders is risk neutral, so they will locate in both buy and sell side with rate $\frac{1}{2}(1 - \mu)\delta$. If the expected price equals $S_t \leq \underline{S} = S_T$ informed Traders will arrive at buy side with probability $(1 - \delta)\mu$ and uninformed traders at rate $\frac{1}{2}(1 - \mu)(1 - \delta)$.

Whilst forward knowledge of the fix has many applications it is worth postulating a specific example of a trading strategy in ED futures around LIBOR announcements. Consider the period 2005 to 2007, the spread of submissions for the LIBOR fix are very tight. For a set

of banks typically sitting in the middle eight submitters, a collaborative submission (as indicated by the cross bank emails noted previously) could allow for reasonable fore-knowledge of the fix in the second decimal, for instance an unexpected increase in the submission by one basis point or \$25 for each contract. The per day quote volume (in terms of numbers of contracts quoted at the inside best bid best ask) for the Eurodollar market exceeds 7¹⁵ volume for a single day. Therefore a \$25 dollar a contract advantage has potential for substantial profits, if a suitable shock can be engineered against the current market expectations. It is worth noting that as a forward rate contract, the shock can be engineered from any LIBOR fixing, however, it is apparent that the size of the impact needs to be reasonably close to maturity contracts.

As the update on the reference rate is at 11:00.00 GMT time we can think of this as being the realization of a signal that will update the information set of all traders. Prior to the update the request to the submitter pool for a ‘high’ or a ‘low’ fixing equates to ‘ δ ’ and ‘ $1 - \delta$ ’ respectively.

Shortly after 11:00:00 GMT , around 11:30:00 GMT or 06:30:00 central time (recalling that the ED future is traded around the clock) the LIBOR rate is announced, then information set for the futures are updated for all participants as a realization. However, in a smaller scale study on ED futures Kim et al. (2014) reports that the highest fraction of informed trading is located in the ED market around 7.00 to 10.00 CT which is after the announcement rather than in the lead up. I will show later that this is consistent with my macro-view argument that maturity and activity play a much more important role in the degree of informed trading and order toxicity within the market. After receiving the signal, informed derivatives traders wait to ‘Buy’ or ‘Sell’ regarding which type of information they have received from the submitter. If they arrive on buying side then $\mathbb{P}(Buy) = 1 + \mu(1 - 2\delta)/2$, while on the sell side $\mathbb{P}(Sell) = (1 - \mu)(1 - 2\delta)/2$, hence $\mathbb{P}(Buy) + \mathbb{P}(Sell) = 1$. Subsequently, in each trading period the nature of the event occurs with, $\mathbb{P}(Buy) = \mathbb{P}(sell) = 0.5$ then

$\delta_t = 0.5, \mathbb{P}(S_T = \underline{S})$ and $\mathbb{P}(S_T = \overline{S}) = 1 - \delta$ at $t = 0$.

In the period after the LIBOR is published, informed traders arrive with a degree of positive adverse selection that can be observed by increased spread. At this point, a market maker recognizes a price level at which he intends to enter, whether long position (Bid, B) or short position (Ask, A). Although the market maker doesn't know that other market participants at period t are informed or not, he can update his belief about the value of S_T as trades are revealed, so $\mathbb{P}(S_T = \underline{S} | Buy) = \mathbb{P}(S_T = \underline{S}, Buy) / \mathbb{P}(Buy) = \delta_t(1 - \mu) / (1 + \mu(1 - 2\delta))$. If $(S_T) = \underline{S} \leq S_t$ informed traders lower their expectation, since $E[S_T | Buy] = \underline{S}(1 - \mu)\delta_t + \overline{S}(1 + \mu)(1 - \delta_t) / (1 - \mu(1 - 2\delta_t))$ and increase expectation on S_T , since $E[S_T | Sell] = \underline{S}(1 + \mu)\delta_t + \overline{S}(1 - \mu)(1 - \delta_t) / (1 + \mu(1 - 2\delta_t))$. Therefore the spread is $A - B = 4(1 - \delta_t)\delta_t\mu(\overline{S} - \underline{S}) / (1 - (1 - 2\delta_t)^2\mu^2)$.

Conversely, in the case of $(S_T) = \underline{S} \leq S_t$ informed trading (μ) occurs when $A = E[S_T | Buy]$ with a profit is gained by direct wealth transfer from the loss incurred by uninformed trades (ε). I can therefore write $(A - E[S_T | \mu, Sell])\mathbb{P}(\mu | Sell) = (A - E[S_T | \varepsilon, Sell])\mathbb{P}(\varepsilon | Sell)$, since by construction $\mathbb{P}(\mu | Sell) + \mathbb{P}(\varepsilon | Sell) = 1$. This argument can be made for B when $(S_T) = S_t \leq S_t$. Recall that the original interpretation of the Glosten and Milgrom (1985) model operates in a zero sum outcome. Definitively, a futures market can be seen in this sense as the settlement at marking to market, from the price at 13:59:00 and 14:00:00 CT (19:59:00 to 20:00:00 GMT), provides a definitive result on the days activities. Therefore, one explanation for the post fixing informed trading is that the update of the LIBOR curve itself generates new differentiated sets of information on $E(S_T)$ between informed traders and uninformed traders prior to the final settlement at 14:00:00.

3.2. The Asymptotic Standard Error of The PIN

The previous discussion indicates why a binomial high versus low fix model is very appropriate for this setting, indeed one could argue the relatively clean mechanism for discerning the days winners and losers makes this far more appropriate than the many other applica-

tions of this approach. The original PIN is a measure by Easley et al. (1996) of asymmetric information fitted via maximum likelihood estimation directly from measurements of the order flow. The tree diagram in Figure 1 presents the dynamics of the game and frames the original PIN type model in terms of the information structure of the ED futures market.

I restrict my interest and the estimation of the model to each contract individually. Whilst attempting to build a simultaneous equation model across all 40 contracts seems attractive there are substantial drawbacks in terms of numerical tractability. This analysis index discrete trading time by $i = 1, 2, \dots, I$ and time is considered to be continuous within discrete trading block and is denoted by $t \in [0, T)$. Market participants buy or sell the asset at bid and ask prices posted on the limit order-book during the trading period. Information events are independently distributed at the beginning of each trading block, and occur with probability α . However, informed traders will only trade when they perceive that their LIBOR fixers will be able to generate an information event; they will buy an asset if they receive a signal of good news with probability $1 - \delta$ and sell if they receive a signal of bad news with probability δ . As ε_b and ε_s are the selling and buying rate of uninformed traders, who are supposed to have the same intensity (0.5), therefore the probability of uninformed traders is $\varepsilon_s = \varepsilon_b = \varepsilon$. In this way, the set of parameters is reduced to α, δ, μ and ε .

Within each trading block, order-flow arrives from both uninformed traders and informed traders subject to the receipt of their signal. Both uninformed buyers and sellers arrive in the market at rate ε . Competitive informed traders who are risk neutral will arrive when information events have occurred. If they receive a definitive signal of an up tick, they will arrive to buy orders; conversely, they will submit sell orders if they receive a bad signal or bad news. The arrival rate for this process is μ . Following convention, the arrival processes are assumed to be independent. On good event days, the arrival rate for buy orders is $\varepsilon + \mu$ and for sell orders is ε . On the other hand, if there is any bad news or bad signal on any trading day, the arrival rate of sell orders is $\varepsilon + \mu$ and buy orders arrive at a rate of μ .

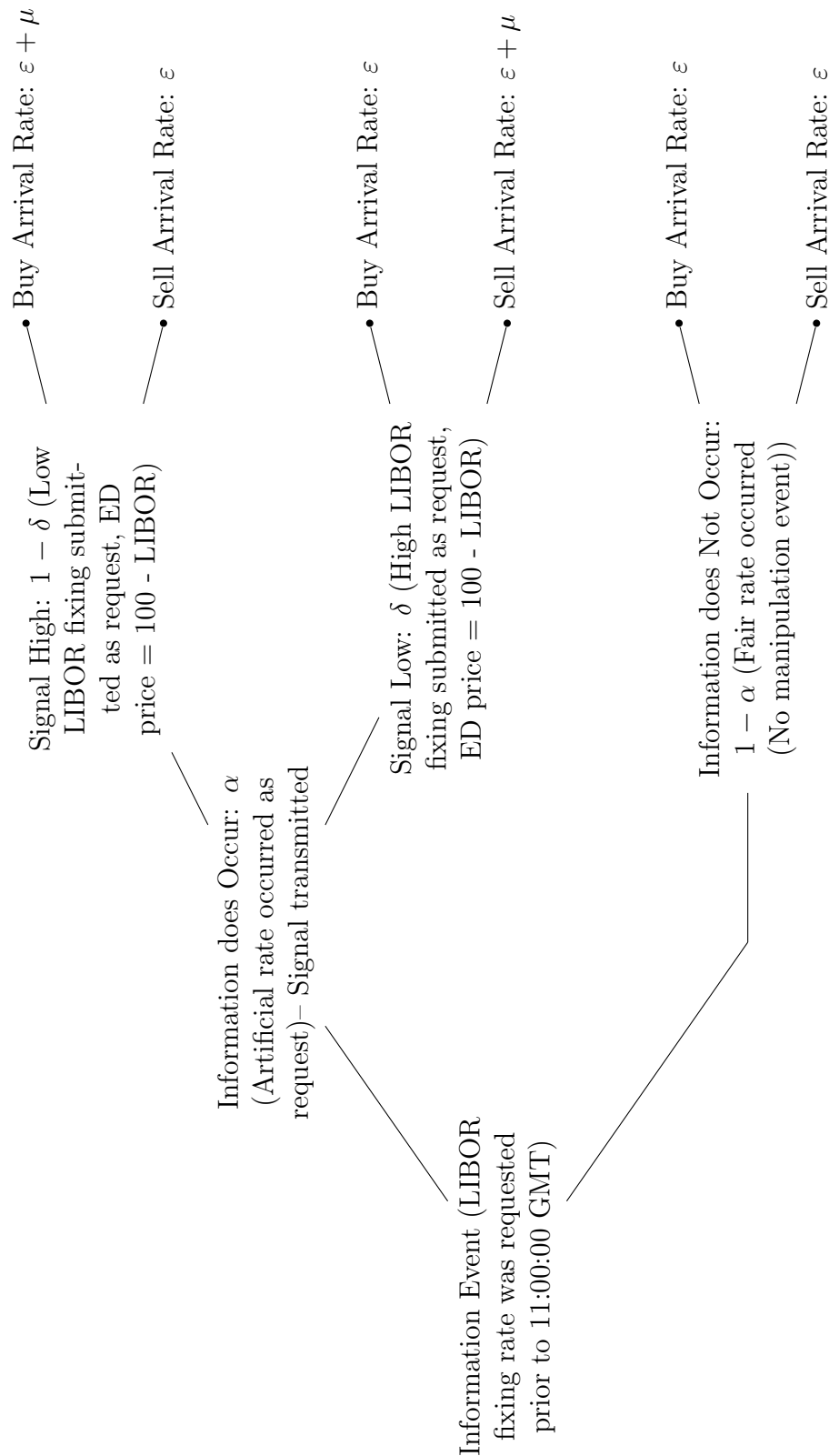


Figure 1: The Eurodollar price and a tree diagram of the sequential trading progress

Note: that after 11.00 GMT the LIBOR FIX is published, if the rate that occurred has been requested by derivatives traders, then the published data is similar to a signal with rate α . When the fix is high, the price of the contract will decrease (ED price = $100 - \text{LIBOR}$) and when the fix is low the price will be higher, if the bank can guarantee that its fix will impact on the LIBOR average determination.

Finally, if there is no news or no signal on that day, only uninformed traders arrive for both buy and sell orders at arrival rate ε .

In each block, the news arrival contains one of three types of information. On the CME Globex market makers provide liquidity via the ‘mass-quote’ system whereby a single market maker can control a very large number of standing quotes within the various levels of the limit order book (effectively mimicking a larger group of individual traders). Following Easley et al. (1996) it can be presumed that the market maker knows that there is some probability attached to each branch and has some knowledge of the order arrival process for each branch. However, the market maker does not know which of the three branches has been selected. Since he cannot directly observe which type of branch has been selected he uses Bayesian updating from the observed order-flow to adjust his beliefs about the nature of the information events during a single trading block. Let $P(t) = (P_n(t), P_b(t), P_g(t))$ be a liquidity provider’s belief about the occurrence of information event “no news” (n), “bad news” (b), and “good news” (g) at time t . Then, his prior beliefs at time 0 is $P(t_0) = (1 - \alpha, \alpha\delta, \alpha(1 - \delta))$.

To determine quotes at time t , the market maker updates his prior belief on the condition of an arrival order of the relevant type. For instance, the bid at time t , $B(t)$, is the expected value of the asset conditional both on the history of the process prior to the arrival of order at time t and on the fact that someone wants to sell the asset. Let (s_t) denote the event that a sell order arrives at time t . Let $P_n(t|s_t)$ be the market maker’s updated belief vector conditional on the history prior to time t and on the event that a sell order arrives at time t .

Following Bayes’ rule, the market maker’s posterior probability on no news at time t , if an order to sell arrives at t , is $\mathbb{P}_n(t|s_t) = \mathbb{P}_n(t) \cdot \varepsilon/\varepsilon + \mathbb{P}_b(t)\mu$, the posterior probability on bad news is $\mathbb{P}_b(t|s_t) = P_b(t) \cdot (\varepsilon + \mu)/\varepsilon + P_b(t)\mu$ and the posterior probability on good news is $\mathbb{P}_g(t|s_t) = \mathbb{P}_g(t) \cdot \varepsilon/\varepsilon + \mathbb{P}_b(t)\mu$. Comparable to the case of buy orders (B_t) the posterior probability on good news at time t is $\mathbb{P}_g(t|b_t) = \mathbb{P}_g(t) \cdot (\varepsilon + \mu)/\varepsilon + \mathbb{P}_g(t)\mu$ the posterior

probability on bad news is $\mathbb{P}_b(t|b_t) = \mathbb{P}_b(t) \cdot \varepsilon/\varepsilon + \mathbb{P}_g(t)\mu$ and the posterior probability on no news is $\mathbb{P}_n(t|b_t) = \mathbb{P}_n(t) \cdot \varepsilon/\varepsilon + \mathbb{P}_g(t)\mu$. At the end of the trading on any day, the full information value of the asset is realized. If it is good news on trading day (i) the informed trader knows that the value of the asset at the end of the day is worth $\bar{S}_i$, similarly it is $\underline{S}_i$ if it is bad news on day i and the asset on day i is worth $S_i^* = \delta\underline{S}_i + (1 + \delta)\bar{S}_i$, (so, $\bar{S}_i > S_i^* > \underline{S}_i$) if there is no news at all.

To complete the bid or ask at time t , the liquidity provider updates his position on the condition of arrival order according to the information type. At time t the expected value of the asset, conditional on the history of trade prior to time t , is $\mathbb{E}[S_i|t] = \mathbb{P}_n(t)S_i^* + \mathbb{P}_b(t)\underline{S}_i + \mathbb{P}_g(t)\bar{S}_i$ where $S_i^* = \delta\underline{S}_i + (1 + \delta)\bar{S}_i$ is the prior expected value of the asset. How might this work in practice on Globex? Well the market maker uses the mass quote system to place a large number of resting quotes at various levels within the order-book. If one side of the order-book receives a disproportionate number of trades then resting quotes on the opposite side (at level one of the limit order book) update and are canceled, and new quotes are instigated to restore balance and maintain the magnitude of the spread. New quotes deeper in the order book on the side with the draw down are instigated automatically. Whilst, in principle, this should ensure that price shifts are driven by supply and demand, the ‘Flash Crash’ of May 2010, originating on the Globex platform, indicates that this mechanism is not always perfectly functional. From this adjustment process the bid is the expected value of the asset conditional on someone wanting to sell the asset to a liquidity provider. Thus, the bid–ask spread at time t is denoted by $\Sigma(t) = A(t) - B(t)$. This spread is $\Sigma(t) = \frac{\mu P_g(t)}{\varepsilon + \mu P_g(t)}(\bar{S}_i - \mathbb{E}[S_i|t]) + \frac{\mu P_b(t)}{\varepsilon + \mu P_b(t)}(\mathbb{E}[S_i|t] - \underline{S}_i)$. The first term in the bid–ask spread equation is a probability of buy order based on information and the second is the term for sells. The spread for the initial quotes in the period, Σ , has a particular simple form in the natural case in which good and bad events are equally likely. That is, if $\delta = 1 - \delta$, then $\Sigma = \frac{\alpha\mu}{\alpha\mu + 2\varepsilon}(\bar{S}_i - \underline{S}_i)$. The key component of this model is the probability that an order is

from an informed trader, which is called the PIN = $\frac{\alpha\mu}{\alpha\mu+2\varepsilon}$, where $\alpha\mu + 2\varepsilon$ is the arrival rate for all orders and $\alpha\mu$ is the arrival rate for information-based orders. Therefore, the PIN is a measure of the fraction of orders that arise from informed traders relative to the overall order flow, and the spread equation shows that it is the key determinant of spreads. These equations clarify that liquidity providers need to correctly estimate the PIN in order to identify the optimal levels at which to enter the market. An unanticipated increase in PIN will result in losses to those liquidity providers who do not adjust their prices.

It is difficult to estimate the parameter vector $\theta = (\alpha, \delta, \varepsilon, \mu)$ because it cannot be directly observed in either the occurrence of information events or the associated arrival of uninformed and informed traders. In fact, in terms of measuring the daily arrival rate of sell (s_t) and buy (b_t) it is possible to infer these values using maximum likelihood and assuming that the trading process follows a Poisson process (Karyampas and Paiardini, 2011).

The Easley et al. (1996) PIN estimator considers the likelihood of order arrivals during a discrete trading block. In this model, the likelihood of observing sell S and buy B orders on each type of information occurs on a no event block with probability $1 - \alpha$, a bad event block with probability $\alpha\delta$ and a good event block with probability $\alpha(1 - \delta)$. Therefore, the likelihood is

$$\begin{aligned} L_i(\theta) = & (1 - \alpha)e^{-\varepsilon_b} \frac{(\varepsilon_b)^{B_i}}{B_i!} e^{-\varepsilon_s} \frac{(\varepsilon_s)^{S_i}}{S_i!} \\ & + \alpha\delta e^{-\varepsilon_b} \frac{(\varepsilon_b)^{B_i}}{B_i!} e^{-\varepsilon_s + \mu} \frac{(\varepsilon_s + \mu)^{S_i}}{S_i!} \\ & + \alpha(1 - \delta)e^{-\varepsilon_b + \mu} \frac{(\varepsilon_b + \mu)^{B_i}}{B_i!} e^{-\varepsilon_s} \frac{(\varepsilon_s)^{S_i}}{S_i!}. \end{aligned}$$

Since only one type of information occurs in each trading block, the maximum likelihood estimator of the information event parameters α and δ will be either 0 or 1. However, these parameters can be estimated from each block of buy and sells, assuming that the blocks are approximately independent. The likelihood of observing the data $M = (B_i, S_i)_{i=1}^I$ across I

trading block is just the product of the daily likelihood function: $L(M|\theta) = \prod_{i=1}^I L(\theta|B_i, S_i)$.

The PIN estimates are computed by maximizing the parameter vector θ from any data set M , which is normally taken to be from the daily trades and quotes. For certain experiments I also use 30, 60 and 90 day PIN estimates, these experiments are described in the following section. As a quantity estimated by maximum likelihood I can make use the ‘delta method’ to compute analytical confidence bounds. First, let the log-likelihood function of the PIN be given by:

$$\begin{aligned}\mathcal{L}[M|\theta] &= \sum_{i=1}^I \log(L_i(\theta)) \\ &= \sum_{i=1}^I \log \left(\frac{-\alpha(\delta-1)\varepsilon_i^{S_i}(\mu+\varepsilon)^{B_i} + \alpha\delta\varepsilon_i^{B_i}(\mu+\varepsilon)^{S_i} - (\alpha-1)\varepsilon_i^{B_i+S_i}}{B_i!S_i!} \right) - 2\varepsilon\end{aligned}\quad (3.1)$$

The analytical gradient, $\nabla \mathcal{L}[M|\theta] = \frac{\partial}{\partial \theta} \sum_{i=1}^I L_i(B_i, S_i|\theta)$, of $\mathcal{L}[M|\theta]$ is given by:

$$\nabla \mathcal{L}[M|\theta] = \sum_{i=1}^I \left(\begin{aligned} &\left(\alpha - \frac{\varepsilon_i^{B_i+S_i}}{(\delta-1)\varepsilon_i^S(\mu+\varepsilon)^B - \delta\varepsilon_i^B(\mu+\varepsilon)^S + \varepsilon_i^{B_i+S_i}} \right)^{-1} \\ &\left(\frac{\varepsilon_i^S((\alpha-1)\varepsilon_i^B - \alpha(\mu+\varepsilon)^B)}{\alpha(\varepsilon_i^S(\mu+\varepsilon)^B - \varepsilon_i^B(\mu+\varepsilon)^S)} + \delta \right)^{-1} \\ &\frac{\varepsilon_i^S(\alpha(\delta-1)(\mu+\varepsilon)^B(B_i\varepsilon + S_i(\mu+\varepsilon)) + (\alpha-1)(B_i+S_i)\varepsilon_i^B(\mu+\varepsilon)) - \alpha\delta\varepsilon_i^B(\mu+\varepsilon)^S(B_i(\mu+\varepsilon) + S_i\varepsilon)}{\varepsilon(\mu+\varepsilon)(\alpha(\delta-1)\varepsilon_i^S(\mu+\varepsilon)^B - \alpha\delta\varepsilon_i^B(\mu+\varepsilon)^S + (\alpha-1)\varepsilon_i^{B_i+S_i})} - 2 \\ &\frac{\alpha(B_i(\delta-1)\varepsilon_i^S(\mu+\varepsilon)^B - \delta S_i\varepsilon_i^B(\mu+\varepsilon)^S)}{(\mu+\varepsilon)(\alpha(\delta-1)\varepsilon_i^S(\mu+\varepsilon)^B - \alpha\delta\varepsilon_i^B(\mu+\varepsilon)^S + (\alpha-1)\varepsilon_i^{B_i+S_i})} \end{aligned} \right) \quad (3.2)$$

Let $\nabla^2 \mathcal{L}[M|\theta]$ be the Hessian matrix of second order derivatives such that $\nabla^2 \mathcal{L}[M|\theta] = \frac{\partial^2}{\partial \theta \partial \theta'} \sum_{i=1}^I L_i(B_i, S_i|\theta)$. For the maximized likelihood, with parameter vector $\hat{\theta}$ the Hessian (Fisher information matrix) is denoted by $\nabla^2 \mathcal{L}[M|\hat{\theta}]$. Let $\nabla P[\theta] = \frac{\partial}{\partial \theta} P[\theta]$ and is derived analytically by:

$$\nabla P[\theta] = \begin{pmatrix} \frac{\mu}{\alpha\mu+2\varepsilon} - \frac{\alpha\mu^2}{(\alpha\mu+2\varepsilon)^2} \\ 0 \\ -\frac{2\alpha\mu}{(\alpha\mu+2\varepsilon)^2} \\ \frac{\alpha}{\alpha\mu+2\varepsilon} - \frac{\alpha^2\mu}{(\alpha\mu+2\varepsilon)^2} \end{pmatrix} \quad (3.3)$$

at the maxima, asymptotically, the PIN estimator is derived from a standard binomial tree and as such the likelihood estimator will obey the standard central limit theorems. Subsequently, let $\bar{\theta}$ be the true parameter estimates, from the standard Gaussian limit theorem the Cramér-Rao bound is given by $\sqrt{I}(\bar{\theta} - \hat{\theta}) \sim \mathcal{N}(0, \nabla^2 \mathcal{L}[M|\hat{\theta}]^{-1})$. From the standard delta method, the variance of the PIN estimates will be given by:

$$\sqrt{I}(P[\bar{\theta}] - P[\hat{\theta}]) \rightarrow^d \mathcal{N}(0, \nabla P[\hat{\theta}]' \nabla^2 \mathcal{L}[M|\hat{\theta}]^{-1} \nabla P[\hat{\theta}])$$

It is relatively straightforward to observe that the variance of the PIN estimates $\nabla P[\hat{\theta}]' \nabla^2 \mathcal{L}[M|\hat{\theta}]^{-1} \nabla P[\hat{\theta}]$ is easily specified as follows. Let $I\hat{\Omega} = \nabla^2 \mathcal{L}[M|\hat{\theta}]^{-1}$, with elements $[\hat{\omega}_{ij}]$ where $i, j \in \{\alpha, \delta, \varepsilon, \mu\}$ be the covariance matrix of the physically estimated parameters, $\{\alpha, \delta, \varepsilon, \mu\}$.¹ To test the significance of shifts in the PIN, I compute cumulative sums of the PIN estimates and use the standard central limit theorem to compute confidence intervals. I will now move on to look at the VPIN, which is computed directly from the moments of the order-flow so I compute the error variance of VPIN via iid. bootstrap resampling.

Proposition 1. *The standard error of the estimated PIN: the asymptotic standard error of the PIN is given by*

$$\widehat{st.dev}(P[\bar{\theta}] - P[\hat{\theta}]) \rightarrow^d \sqrt{\frac{4\hat{\alpha}(\hat{\mu}(2\hat{\varepsilon}(\hat{\varepsilon} - \hat{\alpha})\hat{\omega}_{\hat{\varepsilon}\hat{\mu}} + \hat{\alpha}\hat{\mu}\hat{\omega}_{\hat{\varepsilon}\hat{\varepsilon}} - 2\hat{\mu}\hat{\varepsilon}\hat{\omega}_{\hat{\delta}\hat{\varepsilon}}) + \hat{\alpha}\hat{\varepsilon}^2\hat{\omega}_{\hat{\mu}\hat{\mu}}) + 4\hat{\mu}^2\hat{\varepsilon}^2\hat{\omega}_{\hat{\alpha}\hat{\alpha}}}{(\hat{\alpha}\hat{\mu} + 2\hat{\varepsilon})^4}} \quad (3.4)$$

Proof. Follows from the statements above and a detailed derivation is given in the following

¹Whilst the point estimates of the PIN are commonly used across academic and practitioner implementations the exact identification

section. □

Traditional implementations of PIN (see for instance the Matlab implementation by Paolo Zagaglia on the Mathworks webpage and the SAS standard calculation) invoke an unconstrained optimization to compute the PIN. However, I take advantage of the analytical nature of the gradient function and make full use of a standard Newtonian approach, this is both faster and less prone to an erroneous exit. The matrix of second order derivatives allows us to compute an asymptotic standard error, which I can then use to construct my rolling z-tests to detect the significance of changes in PIN over time.

To compute the asymptotic variance of the PIN from the maximum likelihood estimate, I take advantage of the implicit asymptotic normality of the variance covariance matrix from the estimator and proceed using the standard Delta method approach.

Let B_i and S_i be the observed counts of buys and sells from the order-flow, using my trade classification mechanism. Unfortunately, I do not have a good way of measuring the error covariance matrix on B_i and S_i therefore I have to treat the estimates as non-stochastic in this instance. The log-likelihood of the i observation of B_i and S_i from $i \in \{1, \dots, I\}$, given my model setting is identical to the standard PIN formulation, with parameter vector $\theta = [\alpha, \delta, \varepsilon, \mu]'$, is denoted by

$$\mathfrak{L}_i(\theta) = \log \left(\frac{-\alpha(\delta - 1)\varepsilon^{S_i}(\mu + \varepsilon)^{B_i} + \alpha\delta\varepsilon^{B_i}(\mu + \varepsilon)^{S_i} - (\alpha - 1)\varepsilon^{B_i+S_i}}{B_i!S_i!} \right) - 2\varepsilon \quad (3.5)$$

summing over the I observations $\mathfrak{L}(\theta) = \sum_i \mathfrak{L}_i(\theta)$ and taking the first derivative yields the

analytic gradient:

$$\nabla \mathfrak{L}(\theta) = \sum_i \left[\frac{\left(\alpha - \frac{\varepsilon^{B_i+S_i}}{(\delta-1)\varepsilon^{S_i}(\mu+\varepsilon)^{B_i} - \delta\varepsilon^{B_i}(\mu+\varepsilon)^{S_i} + \varepsilon^{B_i+S_i}} \right)^{-1} \left(\frac{\varepsilon^{S_i} \left((\alpha-1)\varepsilon^{B_i} - \alpha(\mu+\varepsilon)^{B_i} \right)}{\alpha \left(\varepsilon^{S_i}(\mu+\varepsilon)^{B_i} - \varepsilon^{B_i}(\mu+\varepsilon)^{S_i} \right)} + \delta \right)^{-1}}{\varepsilon^{S_i} \left(\alpha(\delta-1)(\mu+\varepsilon)^{B_i} (B_i\varepsilon + S_i(\mu+\varepsilon)) + (\alpha-1)(B_i+S_i)\varepsilon^{B_i}(\mu+\varepsilon) \right) - \alpha\delta\varepsilon^{B_i}(\mu+\varepsilon)^{S_i} (B_i(\mu+\varepsilon) + S_i\varepsilon)}} \frac{\varepsilon(\mu+\varepsilon) \left(\alpha(\delta-1)\varepsilon^{S_i}(\mu+\varepsilon)^{B_i} - \alpha\delta\varepsilon^{B_i}(\mu+\varepsilon)^{S_i} + (\alpha-1)\varepsilon^{B_i+S_i} \right)}{\alpha \left(B_i(\delta-1)\varepsilon^{S_i}(\mu+\varepsilon)^{B_i} - \delta S_i\varepsilon^{B_i}(\mu+\varepsilon)^{S_i} \right)} - 2 \right] \quad (3.6)$$

with analytical second derivative $H = \nabla^2 \mathbb{L}[M|\theta]$, with elements $[h_{ij}]$ where $i, j \in \{\alpha, \delta, \varepsilon, \mu\}$ the unique (upper triangular) elements are given as follows:

$$h_{\alpha\alpha} = \sum_i - \frac{((\delta-1)\varepsilon^{S_i}(\mu+\varepsilon)^{B_i} - \delta\varepsilon^{B_i}(\mu+\varepsilon)^{S_i} + \varepsilon^{B_i+S_i})^2}{\mathcal{A}^2} \quad (3.7)$$

$$h_{\alpha\delta} = \sum_i \frac{\varepsilon^{B_i+S_i} (\varepsilon^{B_i}(\mu+\varepsilon)^{S_i} - \varepsilon^{S_i}(\mu+\varepsilon)^{B_i})}{\mathcal{A}^2} \quad (3.8)$$

$$h_{\delta\varepsilon} = \sum_i \frac{\mu \mathcal{D} \varepsilon^{B_i+S_i-1}}{\mathcal{A}^2(\mu+\varepsilon)} \quad (3.9)$$

$$h_{\varepsilon\mu} = \sum_i \frac{\mathcal{G} \varepsilon^{B_i+S_i}}{\mathcal{A}^2(\mu+\varepsilon)} \quad (3.10)$$

$$h_{\delta\delta} = \sum_i - \frac{\alpha^2 (\varepsilon^{S_i}(\mu+\varepsilon)^{B_i} - \varepsilon^{B_i}(\mu+\varepsilon)^{S_i})^2}{\mathcal{A}^2} \quad (3.11)$$

$$h_{\delta\varepsilon} = \sum_i \frac{\alpha\mu(\mathcal{E}\mathcal{G})\varepsilon^{B_i+S_i-1}}{\mathcal{A}^2(\mu+\varepsilon)} \quad (3.12)$$

$$h_{\delta\mu} = \sum_i - \frac{(\mathcal{E}\mathcal{G})\varepsilon^{B_i+S_i}}{\mathcal{A}^2(\mu+\varepsilon)} \quad (3.13)$$

$$h_{\varepsilon\varepsilon} = \sum_i \frac{\mathcal{A}\mathcal{B} - \frac{\mathcal{F}^2}{\varepsilon^2(\mu+\varepsilon)^2}}{\mathcal{A}^2} \quad (3.14)$$

$$h_{\varepsilon\mu} = \sum_i \frac{\mathcal{C}}{\mathcal{A}^2\varepsilon(\mu+\varepsilon)^2} \quad (3.15)$$

$$h_{\mu\mu} = \sum_i \frac{\frac{(-\alpha)\mathcal{A}\mathcal{D}}{(\mu+\varepsilon)^2} - (\alpha B_i(\delta-1)\varepsilon^{S_i}(\mu+\varepsilon)^{B_i-1} - \alpha\delta S_i\varepsilon^{B_i}(\mu+\varepsilon)^{S_i-1})^2}{\mathcal{A}^2} \quad (3.16)$$

where the functions $\mathcal{A}$, $\mathcal{B}$, $\mathcal{C}$, $\mathcal{D}$, $\mathcal{E}$, $\mathcal{F}$ and $\mathcal{G}$ are defined as follows:

$$\mathcal{A} = \alpha(\delta - 1)\varepsilon^{S_i}(\mu + \varepsilon)^{B_i} - \alpha\delta\varepsilon^{B_i}(\mu + \varepsilon)^{S_i} + (\alpha - 1)\varepsilon^{B_i + S_i} \quad (3.17)$$

$$\mathcal{B} = \alpha(B_i - 1)B_i\delta\varepsilon^{B_i - 2}(\mu + \varepsilon)^{S_i} + 2\alpha B_i\delta S_i\varepsilon^{B_i - 1}(\mu + \varepsilon)^{S_i - 1} \quad (3.18)$$

$$\begin{aligned} & - \alpha(\delta - 1)(S_i - 1)S_i\varepsilon^{S_i - 2}(\mu + \varepsilon)^{B_i} - 2\alpha B_i(\delta - 1)S_i\varepsilon^{S_i - 1}(\mu + \varepsilon)^{B_i - 1} \\ & - \alpha(B_i - 1)B_i(\delta - 1)\varepsilon^{S_i}(\mu + \varepsilon)^{B_i - 2} \\ & + \alpha\delta(S_i - 1)S_i\varepsilon^{B_i}(\mu + \varepsilon)^{S_i - 2} - (\alpha - 1)(B_i + S_i - 1)(B_i + S_i)\varepsilon^{B_i + S_i - 2} \end{aligned}$$

$$\mathcal{C} = \alpha(\mathcal{C}_a + \mathcal{C}_b + \mathcal{C}_c) \quad (3.19)$$

$$\mathcal{C}_a = \delta S_i\varepsilon^{B_i}(\mu + \varepsilon)^{S_i} \left(\varepsilon^{S_i}(\mu S_i + \varepsilon) \left(\alpha(\delta - 1)(\mu + \varepsilon)^{B_i} + (\alpha - 1)\varepsilon^{B_i} \right) - \alpha\delta\varepsilon^{B_i + 1}(\mu + \varepsilon)^{S_i} \right) \quad (3.20)$$

$$\mathcal{C}_b = B_i(\delta - 1)\varepsilon^{S_i}(\mu + \varepsilon)^{B_i} \left(\alpha\delta\varepsilon^{B_i}(\mu + \varepsilon)^{S_i}(\varepsilon - 2\mu S_i) + \varepsilon^{S_i + 1} \left(-\alpha(\delta - 1)(\mu + \varepsilon)^{B_i} - (\alpha - 1)\varepsilon^{B_i} \right) \right) \quad (3.21)$$

$$\mathcal{C}_c = -B_i^2(\delta - 1)\mu\varepsilon^{B_i + S_i}(\mu + \varepsilon)^{B_i} \left((\alpha - 1)\varepsilon^{S_i} - \alpha\delta(\mu + \varepsilon)^{S_i} \right) \quad (3.22)$$

$$\mathcal{D} = B_i(\delta - 1)\varepsilon^{S_i}(\mu + \varepsilon)^{B_i} - \delta S_i\varepsilon^{B_i}(\mu + \varepsilon)^{S_i} \quad (3.23)$$

$$\mathcal{E} = (\alpha - 1)S_i\varepsilon^{B_i} + \alpha(B_i - S_i)(\mu + \varepsilon)^{B_i} \quad (3.24)$$

$$\mathcal{F} = \alpha\delta\varepsilon^{B_i}(\mu + \varepsilon)^{S_i}(B_i(\mu + \varepsilon) + S_i\varepsilon) \quad (3.25)$$

$$- \varepsilon^{S_i} \left(\alpha(\delta - 1)(\mu + \varepsilon)^{B_i}(B_i\varepsilon + S_i(\mu + \varepsilon)) + (\alpha - 1)(B_i + S_i)\varepsilon^{B_i}(\mu + \varepsilon) \right)$$

$$\mathcal{G} = (\mu + \varepsilon)^{S_i} - (\alpha - 1)B_i\varepsilon^{S_i}(\mu + \varepsilon)^{B_i} \quad (3.26)$$

setting $\hat{\theta} = [\hat{\alpha}, \hat{\delta}, \hat{\varepsilon}, \hat{\mu}]'$, to be the numerically evaluated parameters such that $\nabla \mathfrak{L}(\hat{\theta}) = \mathbf{0}_4$.

Where $\mathbf{0}_4$ is a 4×1 vector of zeros. Under mild regularity conditions the maximum of the function $\mathfrak{L}(\theta)$ can be quickly computed via the Newton-Raphson approach.

The estimate of the asymptotic covariance matrix $\hat{\Omega} = \text{cov}[\bar{\theta} - \hat{\theta}]$, where $\bar{\theta}$ is the true parameter vector is computed from the Hessian matrix following Fisher's theorem by $I\hat{\Omega} = H^{-1}$.

3.2.1. The Asymptotic Estimate Variance of PIN

From the theoretical composition the formulation of the estimated PIN is $P[\theta] = \frac{\alpha\mu}{\alpha\mu + 2\varepsilon}$. Inserting my estimated values yields $P[\hat{\theta}] = \frac{\hat{\alpha}\hat{\mu}}{\hat{\alpha}\hat{\mu} + 2\hat{\varepsilon}}$. My presumption is that $\sqrt{I}(P[\bar{\theta}] - P[\hat{\theta}]) \rightarrow^d \mathcal{N}(0, \varsigma^2)$, where ς^2 is an asymptotic variance. I can confirm this presumption relatively easily, first by confirming that the gradient $\nabla P[\theta]$, is a smooth function of the

underlying parameters:

$$\nabla P[\theta] = \begin{pmatrix} \frac{\mu}{\alpha\mu+2\varepsilon} - \frac{\alpha\mu^2}{(\alpha\mu+2\varepsilon)^2} \\ 0 \\ -\frac{2\alpha\mu}{(\alpha\mu+2\varepsilon)^2} \\ \frac{\alpha}{\alpha\mu+2\varepsilon} - \frac{\alpha^2\mu}{(\alpha\mu+2\varepsilon)^2} \end{pmatrix} \quad (3.27)$$

and second I can derive the exact formulation via the Delta method: $\varsigma^2 = \nabla P[\hat{\theta}]' \hat{\Omega} \nabla P[\hat{\theta}]$. Substitution and rearrangement yields $\sqrt{I}(P[\bar{\theta}] - P[\hat{\theta}]) \rightarrow^d \mathcal{N}(0, \nabla P[\hat{\theta}]' \nabla^2 \mathbb{L}[M|\hat{\theta}]^{-1} \nabla P[\hat{\theta}])$. Evaluating the term $\nabla P[\hat{\theta}]' \nabla^2 \mathbb{L}[M|\hat{\theta}]^{-1} \nabla P[\hat{\theta}]$, dividing by I and taking the square root, yields the standard error in the proposition 1

3.3. The implementation of Volume-Synchronized Probability of Informed Trading (VPIN)

This work uses a variety of choices for implementing VPIN with arrival rate of informed and uninformed traders in Realized Volatility(RV) style. For original work Easley et al. (2011), the fixed number of volume buckets $VPIN(n)$ is fixed at 50, however I consider $n = 20, 50, 100, 200$ and compute the bucket size (V_i) as a fraction of daily trading volume to avoid the bias of activity or inactivity of trading period. So, the VPIN is calculated with $V_i = 1/20, 1/50, 1/100$, and $1/200$. My choices for complimenting VPIN not only calculate with different n and V_i but also using tick-by-tick data instead of one-minute interval as in the original work. I work with tick-by-tick data because I want to implement this approach with the real high frequency world, trading at the speed of light or in a fraction of millisecond, to investigate how the VPIN performs when it is applied with HFTs data.

Prior implementations of VPIN have utilized it as a separate pricing factor within a standard Fama and Macbeth style cross sectional asset pricing model. Given that the purpose of this chapter is to look specifically at the time series variation in the PIN and VPIN measures around well understood events in the life-cycle of the contract then I need to deal specifically with identification of the variation in the measures. For the PIN I can

appeal to the asymptotic properties of the maximum likelihood estimator to build a standard error; however the suggested calculation of the VPIN does not lend itself to direct derivation of the input parameters Cramér-Rao bound and hence a standard error. I therefore choose a double block bootstrap to compute the confidence intervals. It should be noted that consistency of a bootstrap in this case is effectively a tautology, in that I impose consistency on the estimator as each individual step is known to be consistent, but the actual order-flow imbalance does not have a structural meaning from deep model parameters (although one could reconstruct a model if necessary).

It can be seen that from the delta method approach used to compute the error variance for the PIN it is quite possible that the likely data generating process (from the original PIN type model) results in a VPIN that is not a pivotal statistic (i.e. the distribution of the VPIN statistic is dependent on the true parameters of the model); as such I iid the bootstrap in both the major steps in the calculation, the construction of the volume buckets and the calculation of the resulting VPIN, to compute the confidence intervals for the point estimates.

My suggested two stage nested bootstrap requires 9,801 resampled proceeds as follows:

Step 1: Bootstrap across buckets. This is the outer loop. Let V , P , P^A and P^B be the matrices of volumes, transacted price and nearest preceding bid and ask quote prices over a day with $V^\dagger$, $P^\dagger$, $P^{A\dagger}$ and $P^{B\dagger}$ being a draw from their equivalent i.i.d. bootstrap with replacement counterparts. For my purposes I assume that the rows are jointly bootstrapped such that the resorted rows for V^{**} , $P^\dagger$, $P^{A\dagger}$ and $P^{B\dagger}$ are all identical. As the trade classifications for matched sets of the bootstrapped matrices is identical. The calculation proceeds exactly, with $V_i^\dagger$ and $P_i^\dagger$ replacing V_i and P_i .

Step 2: For each $V_i^\dagger$ and $P_i^\dagger$ compute the bootstrapped $V_\tau^{B\dagger}$ and $V_\tau^{S\dagger}$ and construct the collection of bootstrap order imbalances $OI^\dagger = [V_\tau^{S**} - V_\tau^{B\dagger}]_{\tau \in \{1, \dots, N\}}$.

Step 3: I now add an interior loop. For each $OI^\dagger = [V_\tau^{S^\dagger} - V_\tau^{B^\dagger}]_{\tau \in \{1, \dots, N\}}$ iid resample with replacement 99 times $OI^\dagger$, with each bootstrap resample labelled $OI^\ddagger$ and compute $VPIN^\ddagger$. Retain the median $VPIN^\ddagger$ and return this as $VPIN^\dagger$.

Step 4: Sort the collection of 99 outer loop resampled median $VPIN^\dagger$ and then collect the desired confidence intervals (in my case I choose a 95% confidence interval).

These confidence bounds are applied in the subsequent plots for VPIN presented in subsequent sections and the cumulative error is computed for the confidence bounds in the tabulated VPIN adjustments.

4. Data Sources

The data used in this research are introduced in this section, together with a preliminary processing of Eurodollar Futures data. As previously stated, Eurodollar futures contracts are traded on the CME's Globex platform and CME pit (open-outcry) trades. Data for both electronic and open-outcry are directly obtained from the CME tapes for the period January 1, 1996 to January 1, 2014. Pit trades are quotes from CME by the ED code and GE for the Globex code; the amalgamated tapes are classified under the ED moniker. The volume ratio between the Globex and open-outcry is between three and four orders of magnitude over the sample, so separating the pit trades from the electronic trades currently appears to be less interesting. The CME tapes data for the 44 Eurodollar futures contracts is available from the Thomson Reuter database via the Tick History system. I have conducted an analysis on the 4 monthly contracts, however the data is very sparse and the volumes are very small (up to 5 orders of magnitude for busy versus busy days) compared to the 40 quarterly contracts, so for brevity this is excluded from the results. The data records for 40 quarterly Eurodollar future contracts is separated into two different time-frames, the first of which was from January 1, 1996 to July 31, 2007, and the second, from August 1, 2007 to January 1, 2014.

4.1. The descriptive of the term-structure of PIN and PIN's parameters

Table 1 presents an estimated PIN and Tables 2, 3, 4, 5 present PIN's parameters with its standard deviation for ten to one year to maturity on the period of 1996 until 2013. The results in this table show that the probability of information event between 1996 and 2001 is high, then begins to drop around 2002; the highest of α is in 2000 at 0.968 and 0.933 for ten and one year to maturity, respectively. This trend may be consistent with the PIN as both parameters have a similar trend, however, the highest PIN appears in 1997 with 0.844 and 0.830 for ten and one year to maturity, respectively. During the same period the arrival rate of informed traders and uninformed traders μ, ε fluctuates mostly between 0.4 and 0.6, except for 2000 when ε fluctuates around 0.2. After an upward trend of information based trading for Eurodollar trading in the first period, the downward trend appears between 2002 and 2003. This trend can be investigated from the decreases of α and PIN. After 2003, the PIN and the α are increasing, then turning to a downward trend around 2007. The highest α appears in 2007 at 0.945 for ten year to maturity, however, the PIN is just about 0.631, which is lower than 2004, 2005 and 2006 for the same ten year to maturity. From these results, focusing on PIN and α , this analysis finds that the PIN is not purely driven by α or the probability of information that occurs on the trading day. The low(high) of α does not follow the low(high) of PIN, however, they have a similar trend but are slightly overlapping for some period(I leave a full investigation for future research). Interestingly, after 2008, there is a small variation for all parameters, for instance from 2009 to 2013 the α for ten to one year to maturity shows an average value at 0.4, the average estimate of δ, μ and ε are 0.5. Finally, the average estimated PIN from 2009 has a small variation around 15% from the identical PIN at 0.667.

Table 1: The Term Structure of PIN Coefficient
Time To Maturity

Year	10	9	8	7	6	5	4	3	2	1
1996	0.8098 (0.2788)	0.7647 (0.2848)	0.6914 (0.2887)	0.7150 (0.3087)	0.7153 (0.3015)	0.6993 (0.3263)	0.7623 (0.3099)	0.7713 (0.3043)	0.8092 (0.2780)	0.8170 (0.2772)
1997	0.8444 (0.2517)	0.8103 (0.2577)	0.7483 (0.2728)	0.6866 (0.2718)	0.6975 (0.2525)	0.7456 (0.2917)	0.7374 (0.3073)	0.7836 (0.2879)	0.8056 (0.2716)	0.8300 (0.2646)
1998	0.8065 (0.2724)	0.8068 (0.2699)	0.7802 (0.2813)	0.7366 (0.2694)	0.7154 (0.2653)	0.6922 (0.2787)	0.7360 (0.2425)	0.6876 (0.3036)	0.7278 (0.2999)	0.8073 (0.2717)
1999	0.7276 (0.3015)	0.7817 (0.3071)	0.8105 (0.2893)	0.7850 (0.2922)	0.7607 (0.3121)	0.7632 (0.2721)	0.6647 (0.1925)	0.7087 (0.2421)	0.6933 (0.2554)	0.7109 (0.2823)
2000	0.6997 (0.2434)	0.7331 (0.2447)	0.7671 (0.2464)	0.7560 (0.2546)	0.7698 (0.2731)	0.7546 (0.2654)	0.7360 (0.2770)	0.6667 (0.2073)	0.6778 (0.2194)	0.6757 (0.2310)
2001	0.6388 (0.2132)	0.6522 (0.2229)	0.6796 (0.2537)	0.7133 (0.2747)	0.7203 (0.2759)	0.7458 (0.2829)	0.7090 (0.2502)	0.7106 (0.2578)	0.6832 (0.2259)	0.6962 (0.2019)
2002	0.6381 (0.1838)	0.6073 (0.2027)	0.6251 (0.2351)	0.6698 (0.2567)	0.6834 (0.2875)	0.7152 (0.2731)	0.7634 (0.2852)	0.7364 (0.2538)	0.7050 (0.2418)	0.6215 (0.1799)
2003	0.6167 (0.1817)	0.6161 (0.1629)	0.5717 (0.1830)	0.5974 (0.2061)	0.5957 (0.2002)	0.6568 (0.3044)	0.6989 (0.2989)	0.7195 (0.2949)	0.7078 (0.2601)	0.6927 (0.1959)
2004	0.7098 (0.1951)	0.6468 (0.1662)	0.6279 (0.1625)	0.5763 (0.2106)	0.5526 (0.2160)	0.5789 (0.2618)	0.6672 (0.3165)	0.7043 (0.3137)	0.7282 (0.3069)	0.7128 (0.2752)
2005	0.7122 (0.2551)	0.7101 (0.2464)	0.6849 (0.1237)	0.6645 (0.1463)	0.6748 (0.1658)	0.5695 (0.2214)	0.5816 (0.2352)	0.6060 (0.2856)	0.6807 (0.2944)	0.7503 (0.2958)
2006	0.7046 (0.3113)	0.6888 (0.2662)	0.6922 (0.2590)	0.6922 (0.1300)	0.6852 (0.1345)	0.6705 (0.1247)	0.6431 (0.1588)	0.5489 (0.2764)	0.5885 (0.2989)	0.6941 (0.3212)
2007	0.6314 (0.2956)	0.6687 (0.3046)	0.6797 (0.3022)	0.7076 (0.2549)	0.6874 (0.1736)	0.6798 (0.1300)	0.6627 (0.1297)	0.6588 (0.1001)	0.6383 (0.1703)	0.5902 (0.2684)
2008	0.6338 (0.1935)	0.6585 (0.2315)	0.6612 (0.2903)	0.7146 (0.2832)	0.7142 (0.2702)	0.6843 (0.2283)	0.6794 (0.1043)	0.6721 (0.1254)	0.6572 (0.1352)	0.6289 (0.1870)
2009	0.6692 (0.1006)	0.6450 (0.1502)	0.6496 (0.1805)	0.6510 (0.2376)	0.6975 (0.2376)	0.6924 (0.2342)	0.6960 (0.2073)	0.6840 (0.1017)	0.6696 (0.1210)	0.6584 (0.1311)
2010	0.6782 (0.1149)	0.6685 (0.1226)	0.6728 (0.1025)	0.6480 (0.1327)	0.6599 (0.1275)	0.6653 (0.1631)	0.6960 (0.2048)	0.7054 (0.1713)	0.6845 (0.1281)	0.6869 (0.1163)
2011	0.6884 (0.1048)	0.6828 (0.1214)	0.6839 (0.1164)	0.6806 (0.0979)	0.6634 (0.1212)	0.6340 (0.1998)	0.6245 (0.2392)	0.6146 (0.2971)	0.6731 (0.2308)	0.6814 (0.1194)
2012	0.6976 (0.1190)	0.6961 (0.1153)	0.6776 (0.1101)	0.6845 (0.0987)	0.6676 (0.1206)	0.6721 (0.1282)	0.6552 (0.1356)	0.6520 (0.1747)	0.6293 (0.2129)	0.7058 (0.1735)
2013	0.6866 (0.1401)	0.6876 (0.1379)	0.6854 (0.1041)	0.6784 (0.0853)	0.6825 (0.1086)	0.6705 (0.1200)	0.6811 (0.0984)	0.6593 (0.1177)	0.6537 (0.1356)	0.6341 (0.1788)

Note: This table present the term structure of PIN between 1996 until 2013.

Table 2: The Term Structure of α

Year	Time To Maturity									
	10	9	8	7	6	5	4	3	2	1
1996	0.6862 (0.4172)	0.4397 (0.4599)	0.3694 (0.4425)	0.6110 (0.4523)	0.5798 (0.4530)	0.8135 (0.3238)	0.7971 (0.3365)	0.8427 (0.3084)	0.8444 (0.3012)	0.8241 (0.3215)
1997	0.8397 (0.3043)	0.6715 (0.4132)	0.4508 (0.4367)	0.3725 (0.4275)	0.3328 (0.4167)	0.5229 (0.4702)	0.7255 (0.3800)	0.7540 (0.3791)	0.8000 (0.3468)	0.8392 (0.3130)
1998	0.8959 (0.2522)	0.8844 (0.2619)	0.7302 (0.3890)	0.4949 (0.4274)	0.5258 (0.4232)	0.5335 (0.3774)	0.3227 (0.4406)	0.8212 (0.3408)	0.8005 (0.3570)	0.8366 (0.3305)
1999	0.9298 (0.2191)	0.9261 (0.2156)	0.9233 (0.2249)	0.8853 (0.2477)	0.8506 (0.3224)	0.6524 (0.4398)	0.4157 (0.3204)	0.3710 (0.4701)	0.5723 (0.4867)	0.9254 (0.2356)
2000	0.9541 (0.1925)	0.9620 (0.1614)	0.9612 (0.1583)	0.9594 (0.1481)	0.9556 (0.1587)	0.8109 (0.3394)	0.6672 (0.4331)	0.3857 (0.3629)	0.4373 (0.4262)	0.9333 (0.2304)
2001	0.9683 (0.1575)	0.9851 (0.0972)	0.9643 (0.1633)	0.9645 (0.1554)	0.9612 (0.1622)	0.9740 (0.1090)	0.8149 (0.2865)	0.6860 (0.4060)	0.6992 (0.3771)	0.5722 (0.4806)
2002	0.6755 (0.3670)	0.9828 (0.1190)	0.9677 (0.1621)	0.9571 (0.1746)	0.9569 (0.1713)	0.9419 (0.1761)	0.9209 (0.2088)	0.8182 (0.2702)	0.7052 (0.3065)	0.5992 (0.3175)
2003	0.5988 (0.3169)	0.7065 (0.2925)	0.9769 (0.1379)	0.9909 (0.0848)	0.9904 (0.0767)	0.9624 (0.1627)	0.9572 (0.1533)	0.9390 (0.1779)	0.7771 (0.2984)	0.6109 (0.2919)
2004	0.5980 (0.2987)	0.5023 (0.2621)	0.5978 (0.2893)	0.7392 (0.3532)	0.8601 (0.2875)	0.9028 (0.2731)	0.9586 (0.1633)	0.9374 (0.1996)	0.9291 (0.2053)	0.7428 (0.3145)
2005	0.7553 (0.3012)	0.7535 (0.2942)	0.4468 (0.1807)	0.4667 (0.2188)	0.4357 (0.2498)	0.6762 (0.3245)	0.8120 (0.3071)	0.9315 (0.2232)	0.9634 (0.1441)	0.9427 (0.1769)
2006	0.9342 (0.1766)	0.7797 (0.3052)	0.7320 (0.3152)	0.4393 (0.1761)	0.4435 (0.1789)	0.4517 (0.1855)	0.4578 (0.1988)	0.6563 (0.3587)	0.7772 (0.3337)	0.9482 (0.1631)
2007	0.9457 (0.1843)	0.9563 (0.1595)	0.9019 (0.2247)	0.7725 (0.3003)	0.5291 (0.2530)	0.4455 (0.1837)	0.4720 (0.1658)	0.4978 (0.1389)	0.5703 (0.2200)	0.8735 (0.2680)
2008	0.6584 (0.2726)	0.6865 (0.3322)	0.9094 (0.2159)	0.9414 (0.1759)	0.8248 (0.2727)	0.6707 (0.2991)	0.4517 (0.1622)	0.4701 (0.1836)	0.4811 (0.1698)	0.4864 (0.2538)
2009	0.4845 (0.1628)	0.4891 (0.1991)	0.5514 (0.2519)	0.6466 (0.3087)	0.6862 (0.3077)	0.6782 (0.3016)	0.6019 (0.2790)	0.4497 (0.1613)	0.4627 (0.1515)	0.4806 (0.1536)
2010	0.4397 (0.1827)	0.4598 (0.1814)	0.4721 (0.1708)	0.4822 (0.1773)	0.4741 (0.1885)	0.4311 (0.2395)	0.3419 (0.3009)	0.4302 (0.2676)	0.4533 (0.1695)	0.4363 (0.1780)
2011	0.4585 (0.1481)	0.4399 (0.1749)	0.4397 (0.1829)	0.4453 (0.1781)	0.4638 (0.1924)	0.4619 (0.2192)	0.4168 (0.2958)	0.4388 (0.3440)	0.4217 (0.3001)	0.4484 (0.1528)
2012	0.4462 (0.1588)	0.4565 (0.1410)	0.4628 (0.1403)	0.4675 (0.1315)	0.4564 (0.1445)	0.4290 (0.1882)	0.4592 (0.1667)	0.4096 (0.2393)	0.4162 (0.2548)	0.4631 (0.2430)
2013	0.4589 (0.1892)	0.4524 (0.1789)	0.4718 (0.1279)	0.4829 (0.1188)	0.4743 (0.1395)	0.4621 (0.1522)	0.4496 (0.1580)	0.4763 (0.1751)	0.4741 (0.1764)	0.4715 (0.1950)

Note: This table present the term structure of α between 1996 until 2013.

Table 3: The Term Structure of δ

Year	Time To Maturity									
	10	9	8	7	6	5	4	3	2	1
1996	0.4535 (0.3907)	0.5493 (0.3529)	0.6443 (0.3358)	0.5895 (0.3858)	0.5895 (0.3750)	0.4178 (0.3983)	0.4014 (0.3926)	0.3395 (0.3903)	0.3288 (0.3832)	0.3233 (0.3732)
1997	0.3127 (0.3632)	0.4052 (0.3633)	0.5780 (0.3319)	0.6575 (0.3181)	0.6946 (0.2871)	0.5805 (0.3645)	0.4662 (0.3889)	0.4160 (0.3966)	0.3529 (0.3837)	0.3434 (0.3893)
1998	0.2308 (0.3472)	0.2495 (0.3475)	0.3900 (0.3758)	0.5104 (0.3225)	0.4949 (0.3339)	0.5785 (0.3093)	0.6735 (0.3105)	0.3481 (0.4010)	0.3738 (0.4062)	0.2973 (0.3888)
1999	0.1739 (0.3244)	0.2012 (0.3353)	0.2253 (0.3551)	0.2654 (0.3496)	0.3102 (0.3856)	0.4356 (0.3607)	0.5907 (0.2397)	0.6026 (0.3510)	0.4469 (0.3927)	0.1657 (0.3299)
2000	0.1007 (0.2752)	0.0844 (0.2362)	0.1275 (0.2878)	0.1280 (0.2762)	0.1638 (0.3182)	0.3092 (0.3630)	0.4081 (0.3755)	0.5945 (0.2773)	0.5438 (0.3376)	0.1326 (0.3033)
2001	0.0537 (0.2055)	0.0467 (0.1914)	0.0938 (0.2576)	0.1015 (0.2598)	0.1148 (0.2770)	0.1119 (0.2573)	0.3045 (0.3212)	0.4048 (0.3678)	0.3820 (0.3440)	0.4013 (0.3873)
2002	0.3212 (0.3130)	0.0479 (0.1955)	0.0682 (0.2258)	0.1234 (0.2957)	0.1623 (0.3302)	0.1618 (0.3045)	0.1930 (0.3133)	0.3328 (0.3176)	0.4128 (0.2828)	0.4266 (0.2766)
2003	0.4166 (0.2780)	0.3342 (0.2919)	0.0368 (0.1732)	0.0287 (0.1574)	0.0327 (0.1712)	0.1116 (0.2749)	0.1381 (0.2831)	0.1963 (0.3264)	0.3578 (0.3260)	0.4599 (0.2437)
2004	0.4586 (0.2539)	0.4809 (0.2228)	0.4219 (0.2595)	0.2621 (0.3112)	0.1627 (0.2947)	0.1515 (0.3151)	0.1747 (0.3390)	0.2263 (0.3656)	0.2422 (0.3597)	0.3938 (0.3179)
2005	0.3839 (0.3184)	0.4024 (0.3228)	0.5176 (0.1381)	0.5252 (0.1824)	0.5334 (0.1906)	0.3796 (0.3175)	0.2374 (0.3114)	0.1696 (0.3339)	0.1599 (0.3185)	0.1818 (0.3270)
2006	0.2402 (0.3508)	0.3617 (0.3348)	0.3889 (0.3081)	0.5309 (0.1446)	0.5269 (0.1449)	0.5333 (0.1438)	0.5166 (0.1625)	0.3997 (0.3307)	0.3428 (0.3522)	0.2034 (0.3357)
2007	0.1413 (0.3013)	0.1668 (0.3173)	0.2340 (0.3362)	0.3519 (0.3331)	0.4718 (0.1977)	0.5308 (0.1344)	0.5168 (0.1300)	0.5137 (0.1116)	0.4559 (0.1936)	0.2203 (0.3253)
2008	0.4101 (0.2581)	0.3802 (0.3011)	0.2172 (0.3225)	0.1620 (0.2958)	0.3005 (0.3186)	0.4014 (0.2681)	0.5187 (0.1144)	0.5211 (0.1346)	0.5231 (0.1363)	0.5032 (0.1988)
2009	0.5130 (0.1197)	0.5188 (0.1572)	0.4940 (0.1986)	0.4618 (0.2882)	0.4369 (0.2831)	0.4206 (0.2730)	0.4560 (0.2292)	0.5169 (0.1190)	0.5206 (0.1120)	0.5216 (0.1233)
2010	0.5214 (0.1277)	0.5240 (0.1333)	0.5229 (0.1330)	0.5177 (0.1366)	0.5343 (0.1509)	0.5196 (0.1682)	0.5598 (0.2295)	0.5180 (0.1976)	0.5106 (0.1425)	0.5244 (0.1366)
2011	0.5046 (0.1248)	0.5117 (0.1351)	0.5151 (0.1297)	0.5305 (0.1364)	0.5207 (0.1448)	0.5331 (0.1727)	0.5662 (0.2276)	0.5671 (0.2624)	0.5501 (0.2290)	0.5145 (0.1172)
2012	0.5177 (0.1374)	0.5092 (0.1145)	0.5236 (0.1185)	0.5142 (0.1135)	0.5309 (0.1178)	0.5345 (0.1375)	0.5165 (0.1233)	0.5562 (0.1836)	0.5589 (0.1992)	0.5474 (0.1688)
2013	0.5288 (0.1452)	0.5268 (0.1494)	0.5092 (0.1112)	0.5074 (0.0977)	0.5091 (0.1039)	0.5159 (0.1190)	0.5210 (0.1149)	0.5156 (0.1316)	0.5258 (0.1335)	0.5301 (0.1579)

Note: This table present the term structure of δ between 1996 until 2013.

Table 4: The Term Structure of μ

Year	Time To Maturity									
	10	9	8	7	6	5	4	3	2	1
1996	0.6826 (0.3179)	0.6188 (0.3002)	0.5320 (0.2658)	0.5583 (0.2909)	0.5474 (0.2840)	0.5476 (0.3023)	0.5721 (0.2981)	0.6128 (0.3058)	0.6569 (0.3006)	0.6950 (0.3054)
1997	0.7043 (0.2907)	0.6745 (0.2914)	0.6056 (0.2863)	0.5340 (0.2561)	0.5286 (0.2381)	0.5756 (0.2758)	0.5249 (0.2915)	0.5617 (0.2866)	0.6136 (0.2905)	0.6594 (0.2948)
1998	0.6654 (0.2755)	0.6885 (0.2873)	0.6415 (0.3052)	0.5941 (0.2770)	0.5846 (0.2714)	0.5127 (0.2554)	0.5785 (0.2376)	0.5160 (0.2712)	0.5459 (0.2801)	0.6177 (0.2817)
1999	0.6009 (0.2803)	0.6665 (0.3050)	0.7056 (0.3001)	0.6927 (0.3060)	0.6700 (0.3223)	0.6678 (0.2911)	0.5077 (0.1709)	0.5837 (0.2326)	0.5728 (0.2350)	0.5820 (0.2502)
2000	0.5863 (0.2202)	0.6235 (0.2299)	0.6907 (0.2368)	0.7003 (0.2554)	0.7116 (0.2769)	0.6612 (0.2791)	0.6402 (0.2808)	0.5118 (0.1856)	0.5380 (0.2044)	0.5358 (0.1951)
2001	0.5353 (0.1822)	0.5714 (0.2000)	0.6037 (0.2392)	0.6539 (0.2649)	0.6547 (0.2681)	0.6988 (0.2762)	0.6292 (0.2599)	0.6203 (0.2629)	0.5711 (0.2253)	0.5849 (0.1899)
2002	0.5307 (0.1742)	0.5186 (0.1721)	0.5490 (0.2119)	0.5975 (0.2384)	0.6215 (0.2712)	0.6597 (0.2708)	0.6983 (0.2926)	0.6562 (0.2713)	0.6024 (0.2582)	0.4833 (0.1527)
2003	0.4775 (0.1552)	0.4924 (0.1372)	0.4973 (0.1593)	0.5352 (0.1893)	0.5345 (0.1838)	0.6129 (0.2954)	0.6536 (0.2918)	0.6729 (0.2938)	0.6151 (0.2668)	0.5674 (0.2079)
2004	0.5721 (0.2080)	0.4992 (0.1502)	0.4918 (0.1390)	0.4727 (0.1778)	0.4584 (0.1838)	0.4955 (0.2378)	0.6074 (0.3043)	0.6304 (0.3162)	0.6567 (0.3110)	0.6029 (0.2828)
2005	0.6163 (0.2669)	0.6097 (0.2615)	0.5164 (0.1247)	0.5060 (0.1391)	0.5114 (0.1657)	0.4630 (0.1953)	0.4858 (0.2132)	0.5449 (0.2678)	0.6257 (0.2877)	0.6969 (0.2966)
2006	0.6374 (0.3060)	0.5939 (0.2689)	0.5845 (0.2604)	0.5166 (0.1345)	0.5172 (0.1377)	0.5051 (0.1214)	0.4911 (0.1348)	0.4479 (0.2389)	0.5000 (0.2740)	0.6233 (0.3098)
2007	0.5492 (0.2648)	0.5916 (0.2816)	0.5942 (0.2808)	0.6100 (0.2569)	0.5331 (0.1762)	0.5029 (0.1190)	0.5035 (0.1163)	0.5004 (0.0960)	0.4891 (0.1433)	0.5035 (0.2350)
2008	0.5157 (0.1790)	0.5675 (0.2345)	0.6171 (0.2876)	0.6744 (0.2817)	0.6480 (0.2786)	0.5741 (0.2344)	0.5120 (0.1044)	0.5061 (0.1172)	0.4967 (0.1189)	0.4874 (0.1673)
2009	0.5118 (0.1052)	0.4950 (0.1323)	0.5129 (0.1731)	0.5397 (0.2324)	0.5852 (0.2448)	0.5819 (0.2447)	0.5650 (0.2211)	0.5160 (0.1050)	0.5035 (0.1098)	0.4964 (0.1106)
2010	0.5122 (0.1150)	0.5037 (0.1169)	0.5089 (0.1004)	0.4919 (0.1165)	0.5087 (0.1213)	0.5162 (0.1561)	0.5628 (0.2077)	0.5493 (0.1876)	0.5111 (0.1300)	0.5151 (0.1224)
2011	0.5169 (0.1161)	0.5082 (0.1175)	0.5171 (0.1162)	0.5142 (0.1026)	0.5061 (0.1105)	0.4944 (0.1801)	0.5025 (0.2223)	0.5075 (0.2812)	0.5398 (0.2317)	0.5068 (0.1136)
2012	0.5254 (0.1292)	0.5259 (0.1192)	0.5000 (0.1165)	0.5144 (0.1140)	0.4968 (0.1045)	0.5121 (0.1237)	0.4983 (0.1224)	0.5095 (0.1651)	0.4955 (0.1925)	0.5438 (0.1857)
2013	0.5277 (0.1425)	0.5226 (0.1475)	0.5169 (0.1210)	0.5017 (0.0955)	0.5121 (0.1024)	0.5032 (0.1106)	0.5123 (0.1043)	0.4990 (0.1089)	0.4953 (0.1179)	0.4870 (0.1544)

Note: This table present the term structure of μ between 1996 until 2013.

Table 5: The Term Structure of ε

Year	Time To Maturity									
	10	9	8	7	6	5	4	3	2	1
1996	0.4180 (0.2939)	0.3983 (0.2399)	0.4433 (0.2219)	0.4559 (0.3008)	0.5022 (0.2785)	0.5236 (0.3344)	0.4469 (0.3431)	0.4824 (0.3546)	0.4537 (0.3429)	0.4505 (0.3302)
1997	0.4303 (0.3281)	0.4161 (0.3008)	0.4138 (0.2411)	0.4720 (0.2188)	0.4391 (0.1878)	0.4752 (0.2866)	0.4590 (0.3095)	0.4096 (0.3284)	0.4088 (0.3328)	0.4006 (0.3406)
1998	0.4414 (0.3693)	0.4908 (0.3618)	0.4410 (0.3240)	0.4473 (0.2599)	0.4842 (0.2547)	0.4589 (0.2336)	0.4954 (0.1996)	0.5319 (0.3257)	0.5018 (0.3390)	0.4334 (0.3547)
1999	0.5075 (0.3673)	0.4977 (0.3859)	0.5163 (0.3849)	0.5377 (0.3538)	0.5323 (0.3563)	0.5250 (0.2988)	0.5076 (0.1504)	0.5465 (0.2082)	0.5573 (0.2439)	0.5157 (0.3469)
2000	0.5648 (0.3181)	0.5380 (0.3387)	0.6025 (0.3958)	0.6524 (0.3721)	0.6327 (0.3896)	0.5675 (0.3557)	0.5656 (0.3204)	0.4973 (0.1383)	0.5224 (0.1753)	0.5441 (0.2755)
2001	0.6293 (0.2460)	0.6844 (0.2808)	0.6769 (0.3250)	0.6708 (0.3584)	0.6507 (0.3737)	0.6719 (0.3699)	0.6107 (0.3323)	0.5842 (0.3070)	0.5783 (0.2604)	0.5787 (0.1815)
2002	0.6221 (0.1689)	0.6801 (0.2258)	0.6970 (0.2677)	0.6839 (0.3189)	0.6985 (0.3570)	0.6641 (0.3444)	0.5945 (0.3553)	0.5748 (0.3075)	0.5695 (0.2608)	0.5648 (0.1466)
2003	0.5518 (0.1535)	0.6066 (0.1718)	0.7263 (0.1766)	0.7431 (0.2305)	0.7490 (0.2144)	0.7027 (0.3425)	0.6813 (0.3496)	0.6503 (0.3481)	0.5707 (0.3087)	0.5343 (0.2245)
2004	0.5231 (0.2335)	0.5206 (0.1403)	0.5635 (0.1577)	0.6433 (0.1884)	0.6670 (0.1953)	0.6847 (0.2443)	0.7014 (0.3476)	0.6732 (0.3515)	0.6494 (0.3589)	0.5584 (0.3175)
2005	0.5641 (0.3011)	0.5725 (0.2900)	0.4613 (0.1284)	0.5008 (0.1210)	0.5187 (0.1448)	0.6296 (0.1880)	0.6537 (0.2295)	0.7211 (0.3108)	0.6872 (0.3422)	0.6181 (0.3577)
2006	0.6274 (0.3551)	0.6044 (0.3164)	0.5766 (0.3050)	0.4445 (0.1475)	0.4533 (0.1333)	0.4841 (0.1178)	0.5265 (0.1222)	0.6419 (0.2370)	0.6613 (0.2885)	0.6133 (0.3555)
2007	0.6690 (0.3515)	0.6567 (0.3757)	0.6227 (0.3712)	0.5901 (0.3251)	0.4757 (0.2121)	0.4587 (0.1297)	0.5003 (0.1109)	0.5143 (0.0684)	0.5427 (0.1647)	0.6882 (0.2951)
2008	0.6122 (0.2271)	0.6559 (0.2615)	0.7418 (0.3059)	0.7132 (0.3444)	0.6333 (0.3343)	0.5898 (0.2754)	0.4738 (0.0859)	0.4973 (0.0990)	0.5160 (0.0944)	0.5585 (0.1412)
2009	0.5104 (0.0754)	0.5352 (0.1256)	0.5686 (0.1786)	0.6160 (0.2524)	0.5960 (0.2742)	0.5619 (0.2641)	0.5145 (0.2230)	0.4743 (0.1072)	0.4823 (0.0995)	0.5030 (0.1055)
2010	0.4766 (0.0972)	0.4923 (0.1122)	0.4973 (0.0906)	0.5191 (0.0889)	0.5263 (0.0971)	0.5281 (0.1485)	0.5174 (0.1848)	0.4590 (0.1815)	0.4457 (0.1341)	0.4550 (0.1134)
2011	0.4547 (0.1206)	0.4503 (0.1231)	0.4659 (0.1018)	0.4827 (0.0925)	0.5083 (0.0871)	0.5366 (0.1490)	0.5755 (0.1822)	0.5900 (0.2228)	0.5241 (0.2000)	0.4522 (0.1280)
2012	0.4463 (0.1512)	0.4531 (0.1362)	0.4582 (0.1300)	0.4598 (0.1221)	0.4684 (0.1047)	0.5039 (0.1110)	0.5068 (0.1024)	0.5434 (0.1352)	0.5663 (0.1549)	0.4724 (0.1811)
2013	0.4943 (0.1515)	0.4861 (0.1528)	0.4657 (0.1200)	0.4692 (0.1118)	0.4612 (0.1181)	0.4746 (0.1117)	0.4756 (0.0951)	0.5052 (0.0969)	0.5143 (0.0997)	0.5368 (0.1436)

Note: This table present the term structure of ε between 1996 until 2013.

Table 6: Comparison between average ‘Probability of Informed Trading’ estimates across various studies and markets.

Author	Asset	Sample Period	Min–max	Mean
Easley et al. (1996),(PIN)	NYSE – 90 stocks	Oct. 1, 1990 to Dec. 23, 1990	0.120–0.342	0.197
Idier and Nardelli (2011)(PIN)	Euro overnight interbank rate, Money market	Dec. 2000 to Mar. 2008	0.200–0.580	0.480
Easley et al. (2012),(VPIN)	E-mini S&P 500 (CME)	Jan. 1, 2008 to Oct. 30, 2010	0.205–0.830	0.393
	T-Note (CBOT)	Jan. 1, 2008 to Oct. 30, 2010	0.200–0.800	0.401
	EUR/USD(CME)	Jan. 1, 2008 to Oct. 30, 2010	0.150–0.780	0.327
	Brent Crude Oil (ICE)	Jan. 1, 2008 to Oct. 30, 2010	0.200–0.770	0.384
	Silver (COMEX)	Jan. 1, 2008 to Oct. 30, 2010	0.200–0.840	0.411
Abad and Yagüe (2012),(PIN)	Spanish Stock Exchange – 15 stocks	Jan. 1, 2009 to Dec. 31, 2009	0.104–0.501	0.227
Kim et al. (2014),(VPIN)	Intra day trading – Eurodollar Futures (CME)	Jan. 3, 2005 to Dec. 29, 2006	0.760–0.970	0.880
Yan and Zhang (2014),(PIN)	NYSE/AMEX stocks that have data in the ISSM and TAQ databases	Jan. 1, 1983 to Dec. 31, 2004	0.177–0.227	0.201
My PIN	40 Eurodollar Future contracts	Jan. 1, 1996 to Jan 1, 2014	0.369–0.992	0.688
My VPIN	40 Eurodollar Future contracts	Jan. 1, 1996 to Jan 1, 2014	0.045–0.998	0.126

Note: this table compares the PIN and VPIN from this research with others for various products. It can be seen that my PIN is higher than other PINs, which might be affected by the trading mechanism in the futures market which is different from the stock exchange; however, my VPIN is lower than other VPINs.

5. Analysis and Implications: PIN and VPIN comparison

The following section presents various results of PIN measures on individual ED futures contracts following the conditional imputation approach of Easley et al. (1996). It will also present results for VPIN drawing on Easley et al. (2012) approach but with different n and V_i . First, for the average PINs estimated they are in excess 0.5 for the majority of the sample. This is high relative to comparable equity market studies, see Table 6. However, the results coincide with those of Kim et al. (2014) on the CME Globex trades, for a short sample within my sample and Easley et al. (2012) for related US Dollar Treasury notes. Second, the average VPIN from my results is 0.126 which is lower than other estimates; however, the range of VPIN is wider and the minimum VPIN is 0.045 and the maximum is 0.998.

Although from this study the PIN estimates cover between 0.10 and 0.80 in the equities market, in the derivatives market the range is higher with minimum and maximum values

of 0.04 and 0.99, respectively. Also, in the derivatives market the min-max spread is higher than in the equities market, particularly in the interest rate derivatives market. The highest mean of PIN appears in Kim et al. (2014) with a value of 0.88 ;yet, the highest actual value of PIN is shown in this work with the maximum value of 0.992. The minimum VPIN is 0.040 and the maximum is 0.998. Interestingly, the value of PIN in the Eurodollar Futures is higher than the equity market and always high as presented in this work the PIN is mostly higher than 0.50. However, despite my PIN being constantly high, the average VPIN in this study is lower than others such as in Kim et al. (2014) in which it varies between 0.76 and 0.97. Table 6, illustrates this point and this aspect of my study would be interesting to investigate in the future.

5.1. Following the VPIN Dispute

The VPIN as a tool for detecting excessive levels of order imbalance has been subject to considerable academic discussion and it is worth reviewing some of the key input choices in determining its value. As previously noted, Andersen and Bondarenko (2013, 2014c) and Abad et al. (2015) have criticized the VPIN primarily due to the lack of a good mechanism to be able to choose the correct bulk volume classification. Hence the number of buckets (n) and the bucket size (V_i) are nuisance parameters with no simple method of constructing an appropriate statistical loss function to provide guidance on their values. My solution to this problem is to repeat the analysis over a wide range of different n and V_i and, indeed, I will show that the VPIN, at times, can be sensitive to these choices. Furthermore, this work computes the VPIN for VPIN20, VPIN50, VPIN100 and VPIN200, then compares them with original point estimates of the PIN. I can observe from this that there are substantively different values of VPIN during January 19-29 2010; the minimal value of VPIN is VPIN20 and maximal VPIN is VPIN200 almost continuously through this snapshot of data. For example, PIN and VPIN estimates on January 23, 2010, for the PIN is 0.7521 and 0.0132, 0.0096, 0.0096, 0.0048 for VPIN20, VPIN50, VPIN100 and VPIN200, respectively. From

Table 7: Long run correlation coefficients of PIN, VPIN20, VPIN50, VPIN100 and, VPIN200 on ED?0.

	VPIN20	VPIN50	VPIN100	VPIN200	PIN
VPIN20	1				
VPIN50	0.9250 (0.0000)	1			
VPIN100	0.3127 (0.0000)	0.349 (0.0000)	1		
VPIN200	0.2938 (0.0000)	0.3297 (0.0000)	0.2484 (0.0000)	1	
PIN	0.1156 (0.0000)	0.1362 (0.0000)	0.1343 (0.0000)	0.0843 (0.0051)	1

Note: This table illustrates the matrix of correlation between PIN, VPIN20, VPIN50, VPIN100 and, VPIN200 also a matrix of p-values for testing the hypothesis of no correlation with ($p < 0.05$)

this example we can see that the absolute value of VPIN is sensitive to the choice of n and V_i is certainly not in question. However, the correlations and cumulative differences across choices of n and V_i indicate a high level of agreement in direction, if not in level, which is in keeping with the commentary made in Easley et al. (2014).

Table 7 presents the correlation matrix and p-values of VPIN and PIN. Noticeably, there is a positive and significant correlation between all five variables. This indicates that there is a significant relationship between all five variables: the highest correlation is between VPIN20 and VPIN50 with $r = 0.9250, n = 1100, p = 0.000$ this decreases to $r = 0.3127, 0.2938$ and 0.1156 for VPIN100, VPIN200 and PIN respectively.

Figure 2 also illustrates the variability of the VPIN as a function of different choices of n and V_i . The solid line represents the VPIN estimates for March 2002 maturing futures, the dotted line represents VPIN for June 2002 maturing contracts, and the dashed line represents the VPIN for September 2002 contracts. For each contract I have five different

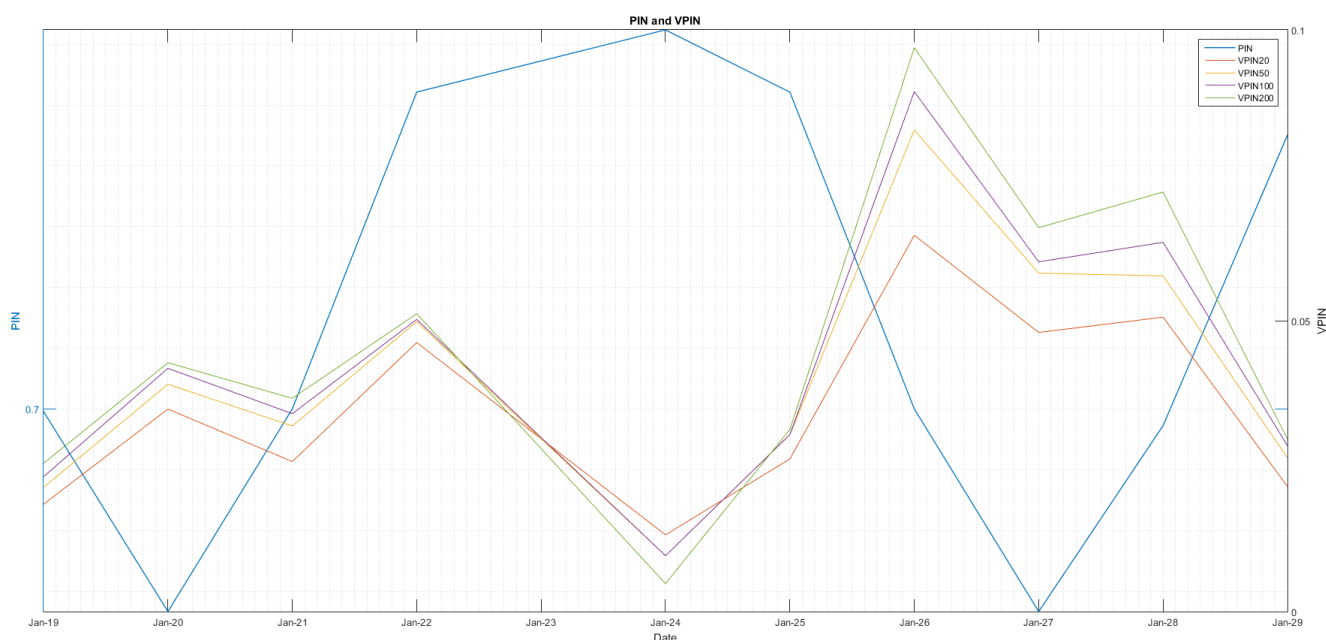


Figure 2: Comparison of a four different types of VPIN on EDH2, EDM2, and EDU2.

Note: This figure presents the compared plots of five different types of VPIN such as VPIN20, VPIN50, VPIN100 and VPIN200 between February 27 until March 09, 2012. It can be seen that VPIN is sensitive to different n and V_i , however there is systematic trend between four types of VPINs. They can be described as, VPIN with $n = 20$ with $V_i = 20$, $n = 50$ with $V_i = 50$, $n = 100$ with $V_i = 100$, and $n = 200$ with $V_i = 200$, also the thick line represents EDH2, the dotted line represents EDM2 and the dash line represents EDU2.

colors that represent different n and V_i choices ($n = 20$ with $V_i = 20$, $n = 50$ with $V_i = 50$, $n = 100$ with $V_i = 100$, and $n = 200$ with $V_i = 200$). The sensitivity, in level, of the VPIN to n and V_i is self-evident however, the cumulative differences are highly consistent.

5.2. Empirical evidence, PIN and VPIN on Eurodollar Futures Market

This section presents the results of cross-sectional PIN and VPIN on the Eurodollar Futures market across 40 contracts from 1996 to 2013 (Figure 3). For the entire sample, I find the PIN and VPIN fluctuate with a downward sloping trend.

The left subplot of Figure 3 shows the average PIN on the left axis, and average VPIN20, VPIN50, VPIN100, and VPIN200 on the right axis. It also shows fluctuations in PIN and VPIN. The highest PIN occurs during 1996 to 1998 with a variation around 0.80-0.82.

It decreases after 1998 with its lowest value at 0.58 around mid 2002. The PIN, after a downward trend from 1996 to mid 2002, bounced back and remains constantly high until 2013, at between 0.065 to 0.070. I assume the high value of PIN between 1996 to mid 2002 is caused by trading activity that operates on an open-outcry platform. This platform, due to its greater information content, has more effect than the electronic market, and can increase the PIN. This has been noted by previous studies on the effect of migration from open-outcry to electronic platforms (Shah and Brorsen (2011); Ates and Wang (2005); Aitken et al. (2004); Tse and Zabolina (2001)). This pattern, albeit with some variation, also appeared on the VPIN.

Overall, the VPIN shows a monotonic decrease from a high point of about 0.40 in 1996 to between 0.01 and 0.02 in 2012. It gradually drops after CME launched the Globex trading system in 2002. However, around 2004, the VPIN bounces back. This pattern appears after routine manipulation of the LIBOR as noted in the FSA and CFTC documents. At the same time, the VPIN is also constantly high although not higher than pre-2002. However, I find a significant decrease in the VPIN around May 2009, when it dropped from about 0.14 to 0.08 compared to the previous period. This decrease is a result of the launch of the LIBOR manipulation investigation. Thus, historical analysis of both PIN and VPIN shows variations in the ED futures market consistent with the FSA and CFTC documents. The same pattern can be observed in PIN and four types of VPIN.

The right subplot of Figure 3 presents cumulative Δ PIN and Δ VPIN. As the PIN and VPIN have differences in their range, it is hard to compare them, so, I had to normalize PIN and VPIN by investigating their differences - ' Δ PIN' and ' Δ VPIN'. Using the same analysis as in the left subplot of Figure 3, I find the same results in the right. That is the major pattern of Δ PIN and Δ VPIN showing a downward trend. Also, the Δ PIN and Δ VPIN had a significant drop after the Globex was launched in 2002, then increased again around 2004. Similarly, I find these two indicators increasing again around the time of the

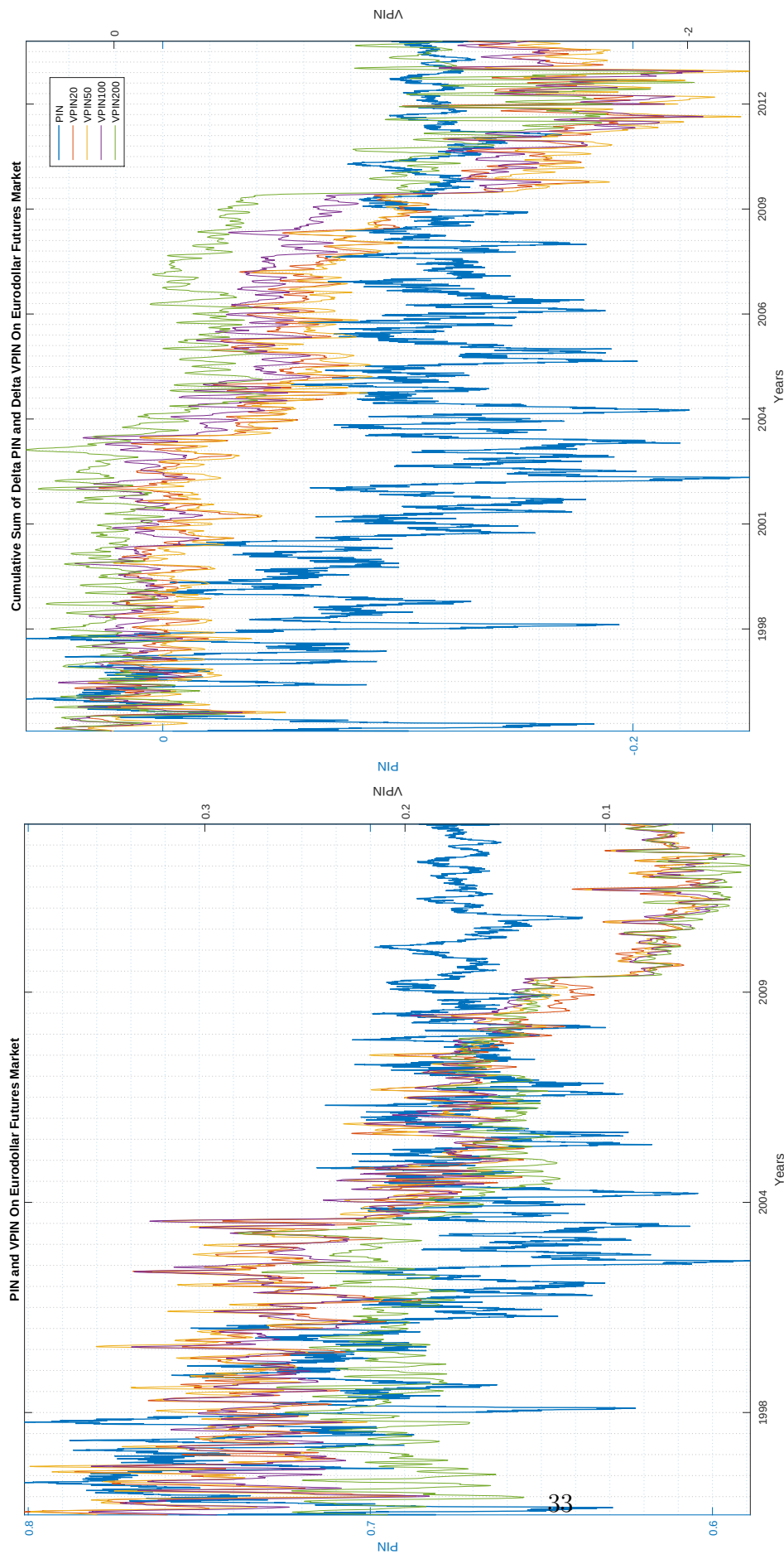


Figure 3: The PIN and VPIN on Eurodollar Futures Market

Note: This Figure presents the PIN and VPIN estimates for the whole cross section of the Eurodollar Futures Market, estimated directly from the limit order book data.

The left subplot of Figure 3, presents the levels of the PIN and VPIN estimates and the right subplot presents the cumulative change in the PIN and VPIN estimates from 1996 to 2013 for ease of interrogation. The VPIN is decomposed into four plots, and each represents a different choice of volume bucket size (20, 50, 100 and 200 contracts, with each contract having a notional value of \$1million).

All of the PIN and VPIN measures have a substantial downward trend, with an approximate 20% decline over the sample. Interestingly, the measure of PIN drops several years prior to the measure of VPIN. It is worth noting that the drop in PIN corresponds to the period just prior to the introduction of the 'Globex' electronic trading platform and then stays relatively low thereafter. The pattern of the trend in VPIN is slightly different, albeit the level of reduction (around a 20% decline from 1996 to 2013) is very similar. The drop off in VPIN occurs in two sharp declines, one in 2003 and a second sharper decline in 2008/9.

financial crisis, then dropping. However, $\Delta VPIN$ appears to be more sensitive than ΔPIN , as at the beginning of 2004 it dropped around 0.5 or 50% compared to 2002. During the same period, ΔPIN dropped only 10%. Also, there was a rapid decrease in $VPIN$ after the LIBOR investigation with $\Delta VPIN$ dropping from -0.05 to -0.15 compared to the previous period.

5.3. Term Structure of PIN and $VPIN$

The left subplot of Figure 4 presents a cross-section of the term structure of PIN for 40 Eurodollar futures contracts for 3,653 days. In this figure I overlay plots of historical PIN against Days-To-Maturity (X axis) to investigate the variation of long term relationships. Although I have 40 different historical PIN s, all show a similar pattern. The figure shows that from 3,653 to 3,000 days to maturity the PIN shows values between 0.62 to 0.92. After this period, the PIN slightly decreases and then drops to its lowest value at 0.32 at around 1,400 days to maturity. However, at this point PIN are highly volatile with a range of 40% with the highest PIN about 0.72 and the lowest 0.32. However, after 365 days to maturity the PIN had a much lower variation of around 0.15 to 0.10. One final finding from this plot is that from 1,400 days to maturity, the term structure of PIN tended to be V-shaped similar to the term structure of the price of futures.

The right subplot of Figure 4 presents a term structure of cumulative ΔPIN for 40 Eurodollar futures contracts for a 10 year period. In this figure, there is a similar pattern in the term structure of PIN and cumulative ΔPIN . Using the same format as in the left subplot of Figure 4 to illustrate the term structure of cumulative ΔPIN , I find a downward trend from the first trading day until 1,800 days to maturity. From the beginning of this trading period the cumulative ΔPIN varies around 20% between 0 to 0.2, then drops to its lowest value of between -0.5 and -0.1. Furthermore, at this point of time the cumulative ΔPIN were extremely volatile with a range of around 40% with a high of -0.08 and a low of -0.52. Furthermore, from around 1,400 days to maturity, the cumulative ΔPIN increases

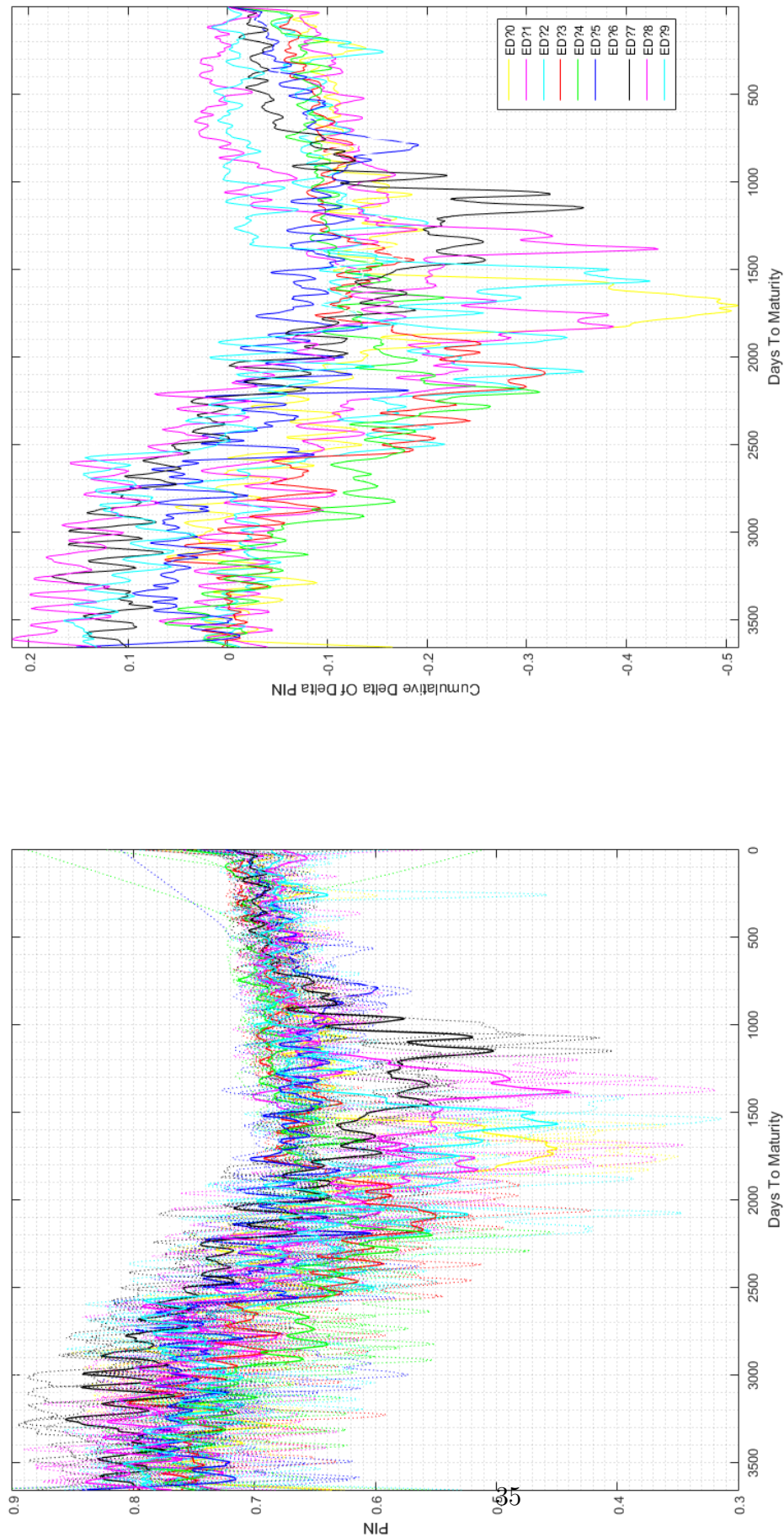


Figure 4: Term Structure of PIN

Note: This figure presents the term structure of PIN that varies between 0.72 to 0.85 from the first trading day to 3,000 days to maturity, then slightly decreases and drops to bottom at around 1,400 days or 3 to 4 years to maturity. The PIN in this period are highly volatile with a wider range compared to the previous period, the highest PIN at 0.72 and the lowest at 0.32. Furthermore, after three years to maturity an upward trend of PIN is shown with less volatility, with the lowest at 0.50 and the highest at 0.75, when it comes to shorter than one year to last trading day.

with less variation which narrows to 10% in comparison to 40% in the last period. Finally, it is interesting to note that the term structure of PIN and cumulative Δ PIN from 1,400 days to expiration have the same V-shape as the typical term structure of futures.

Using the same format analysis from Figure 4, Figure 5 presents the term structure of VPIN on the left subplot and term structure of cumulative Δ VPIN on the right subplot. Although I have shown the term structure of PIN from 3,653 days to expiration, I can present only 1,100 days to maturity for VPIN due to the volume bucket effects. That is to say, as described in the methodology section, for the VPIN algorithm to work, the volumes have to completely fill the volume bucket.

It can be seen from the left subplot of Figure 5 that from 1,100 to 700 days to maturity the term structure of VPIN tends to be U-shaped. The VPIN fluctuates from nearly 0 to just above 0.50 or 50% difference. Later, the variation range decreases from 50% to 20% on 700 days to expiration. Next, between 700 to 400 days to maturity, the VPIN slightly increases from 0.15 to 0.27 respectively, then decreases afterwards. The VPIN drops after 400 days and slightly increases again at around 120 days to expiry. During 400 to 120 days to expiry, the VPIN varies around 10% between 0.10 to 0.20. Interestingly, the differences between the lowest and highest tend to be narrower closer to maturity. This result is consistent with (Balocchi et al. (2001)) in that the range of variation on ED futures price also decreases as the contracts approach expiration. A remarkable result from this plot indicates that from 365 days to expiration, the term structure of VPIN tends to be V-shaped as the term structure of futures prices.

The right subplot of Figure 5 presents the term structure of cumulative Δ VPIN across 40 ED contracts with four different types of VPIN. Overall, the term structure tends to be V-shaped similar to the term structure of futures prices, as the range of variation decreases over the period. The highest Δ VPIN was at the beginning of this period with the minimum at -2.5 and maximum at nearly 1. From 700 days to maturity, the range of variation increases,

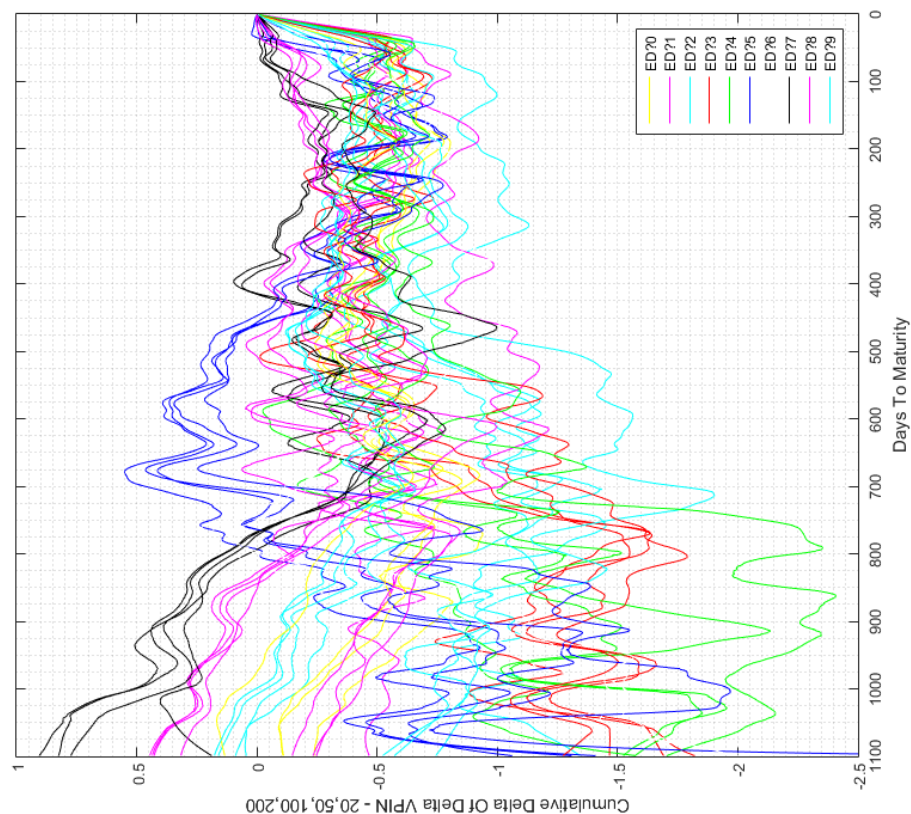
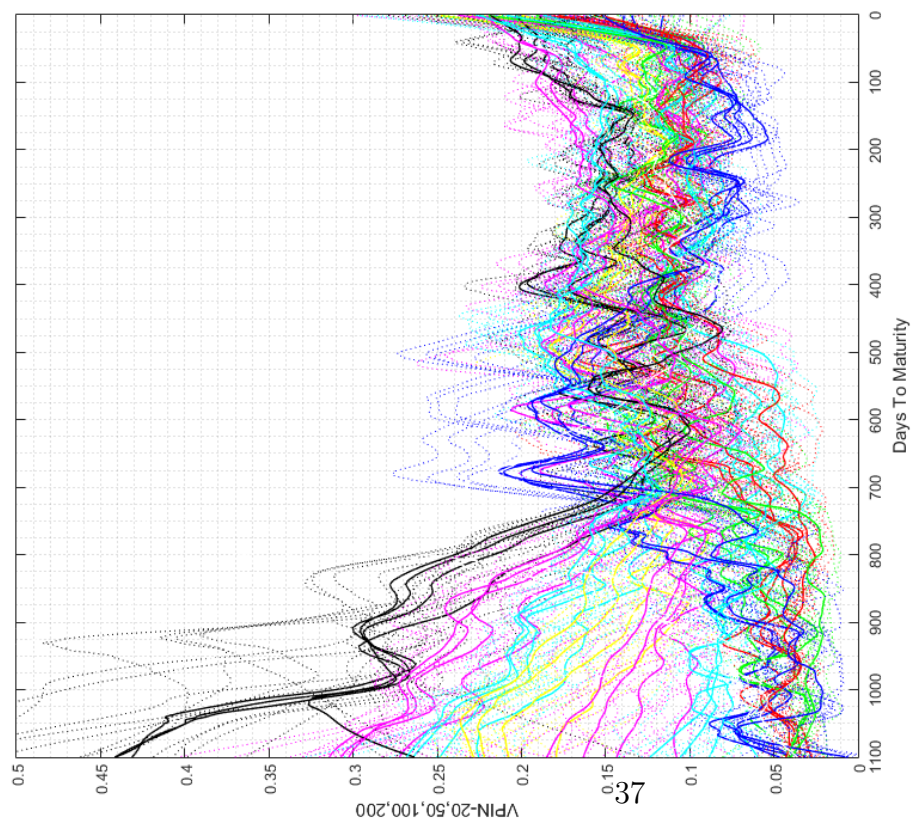


Figure 5: Term Structure of VPIN and Term Structure of $\Delta VPIN$

Note: This Figure presents the term structure of VPIN and term structure of $\Delta VPIN$. Despite the term structure of PIN starting from ten years to maturity, the VPIN can be observed from 1,100 days or around three years to maturity, and this is because there is not enough trading volume per day that can fill the volume bucket on the VPIN calculation. The term structure of VPIN tends to be U-shaped from 1,100 days or around three years to maturity and the variation of VPIN tends to be narrow for a shorter time to maturity. The VPIN drops after 400 days to maturity then has a slight monotonic increase from around 120 days left before the futures contract expires. The result of the narrow variation of VPIN may be consistent with a previous study that the variation of ED futures prices also decreases as the contracts approach expiration (Balocchi et al. (2001)).

which is consistent with the term structure of VPIN when it slightly increases at around 700-days to maturity and decreases afterwards.

For clear comparison, this analysis focuses on PIN and VPIN from 1,100 days to maturity. Therefore, I find a similar pattern of the term structure of ΔPIN and $\Delta VPIN$, which tends to be V-shaped in respect to the term structure of futures contracts. To this end, the behavior of PIN in Figure 4 and VPIN in Figure 5 leads us to investigate whether time to maturity can help to explain the change in the variation of PIN and VPIN, ΔPIN and $\Delta VPIN$. The findings show that time to expiration tends to have an effect on the value and variation of both PIN and VPIN.

5.4. Correlation Surface

The similar patterns of PIN and VPIN against day-to-maturity leads us to investigate their relationship with time to maturity. To investigate this, Pearson's correlation is used to test the correlation between the PIN and four different types of VPIN from 1,000 days to maturity.

Figure 6 presents the correlation between PIN and VPIN20, VPIN50, VPIN100, VPIN200 from 1,000 days to maturity to expiry date. All types of VPIN positively correlate with PIN over time to maturity. However, this varies over the whole period as it increases nearer to maturity. The highest correlation appears between PIN and VPIN20 around 7 days with a correlation of 0.72. Inspecting the relationship between PIN and four VPIN, I find the PIN and the VPIN50 have a higher correlation compared to VPIN20, VPIN100, VPIN200 except after 15 days to maturity when the relationship drops lower than the relationship between PIN and VPIN20. Moreover, around 15 days before the final trading day, the highest correlation changes from VPIN50 to VPIN20 then reaches a peak before the end of the contract. In this period, the lowest correlation appears on VPIN200 followed by VPIN100. This figure pinpoints conclusively a strong effect of time to maturity for PIN and VPIN, as results show a higher correlation nearer to maturity.

Correlation Between PIN, VPIN20, VPIN50, VPIN100, and VPIN200

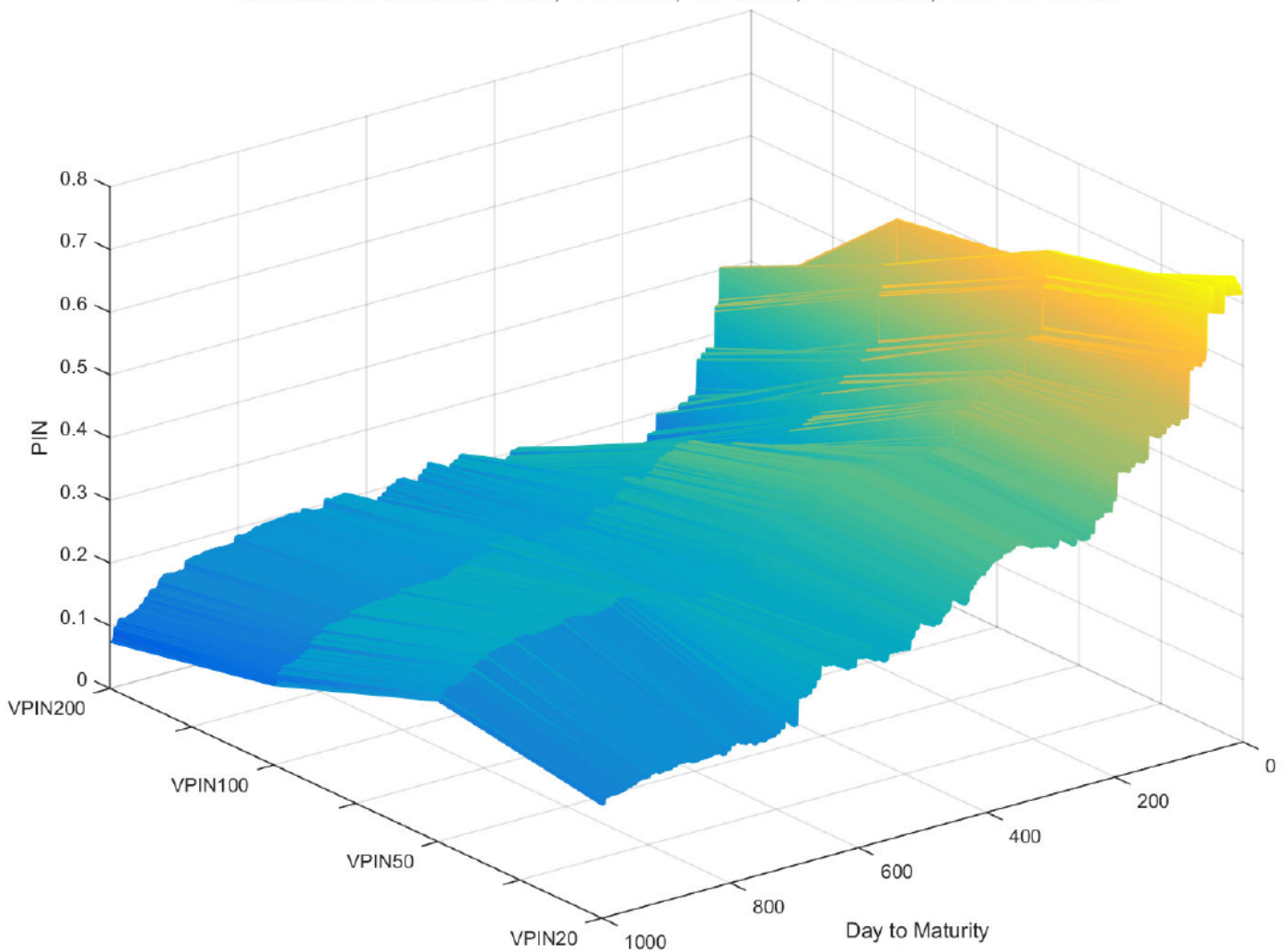


Figure 6: Surface Plot of the correlation surface between the estimated PIN, VPIN20, VPIN50, VPIN100, and VPIN200.

Note: This figure presents the correlation matrix between PIN and VPIN20, VPIN50, VPIN100, VPIN200 from 1,000 days to maturity. It can be seen that the correlation between PIN and VPIN significantly increases for a shorter time to maturity.

5.5. PIN, VPIN and the LIBOR Manipulation

This section presents PIN and VPIN with the LIBOR manipulation case. Most market manipulation studies use general publicly available data. However, I use specific dates of

LIBOR manipulation as cataloged in the CFTC and FSA documents. To further investigate risk surrounding these specific events, I calculate PIN, VPIN20, VPIN50, VPIN100, and VPIN200 to identify short term risk. I also observe long run behavior from 1996 to 2014.

I make the conjecture that LIBOR manipulation has an effect on PIN and VPIN in the futures market as ED trading is based on the LIBOR rate. This analysis begins with an analysis of FSA and CFTC documents regarding dishonest or manipulative submissions of the LIBOR rate issued to major banks, such as Barclays and RBS. The findings indicate that there are two different purposes of manipulation before and after the financial crisis. During the pre-crisis, the banks manipulated the LIBOR, attempting to generate competitive advantage for their trading positions to boost profits. However, during the financial crisis, LIBOR manipulation was an attempt to reduce customer perception of the banks high borrowing costs as this made them look desperate for cash. This was part of a beauty contest designed to ‘reassure’ investors and regulators.

According to the CFTC settlement document, Barclays’ traders attempted to manipulate the US Dollar LIBOR from at least mid-2005 to the autumn of 2007, and thereafter sporadically until 2009. For instance, on March 31, 2006, Barclays employees, including the bank’s senior Treasury managers, sent a request to a US dollar LIBOR submitter to submit a higher rate than normal. The submitter replied he would submit the rate requested.² At this point, I assume that the PIN and VPIN after the manipulation date should be higher than the previous period, then lower afterwards.

The top of the left panel of Figure 7 displays PIN and VPIN around March 31, 2006 as recorded in the LIBOR manipulation document. In this case, I can investigate informed trading around this event by studying the variation of PIN and VPIN by using data from

²CFTC, Order instituting proceedings pursuant to sections 6(c) and 6(d) of the Commodity Exchange Act, as amended, making findings and imposing remedial sanctions; In the Matter of Barclays plc, pp. 9–10.

EDM6 contract, which has the closest expiry date. I investigate PIN and VPIN on EDH6 because this contract is the nearest expiration futures contract with the largest trading volume. Overall, the findings from this analysis show a significant increase in PIN and VPIN. Three days before the event date, the PIN, VPIN20, VPIN50, VPIN100 and VPIN200 move around 0.67, 0.16, 0.16, 0.165 and 0.125 respectively. Then, from the event date the PIN and VPIN sharply increases, achieving a peak on April 02, with the PIN at 0.82, VPIN20 at 0.50 and VPIN50 also at 0.50. However, there is no significant movement on VPIN10 and VPIN200.

For clarification, in the middle row of the left panel, the figure compares Δ PIN, Δ VPIN and the last row presents the PIN and VPIN based on the event date. On the day after the event date, the PIN increases by 0.22 or 22 %, VPIN20 and VPIN50 increases nearly three times when compared with the event date. Furthermore, three days after the event, the PIN and VPIN have significant decrease as the PIN, VPIN20 and VPIN50 shrinks by as much as it increased. Despite PIN, VPIN20 and VPIN50 having a significant movement around the event, VPIN100 and VPIN200 have no significant changes before and after.

The left panel of Figure 7, presents the variation in PIN, VPIN on September 07, 2007, when Rabobank's senior US Dollar trader requested a US Dollar submitter to keep the 3M LIBOR high for the rest of that week.³ This panel shows that between 03 and 13 September 2007, PIN and VPIN have a significant movement characterized by a spike of both on the day after the event. This manipulation influences PIN and VPIN on the EDU7 contract expiring on September 19, 2007. The PIN increases from 0.68, then reaches a peak on September 09 at 0.80, falling to 0.695 on the following day. Also, VPIN20, VPIN50, VPIN100 and VPIN200 increase from around 0.08 before the event to 0.17 on the event date, then reached a peak at 0.29 one day after, as a nearly threefold increase. Finally, similar to the PIN, the

³CFTC, Order instituting proceedings pursuant to detection(c) and 6(d) of the commodity exchange act, as amended, making findings and imposing remedial sanctions in the matter of Coöperative Central Raiffeisen Boerenleenbank B.A., p. 10.

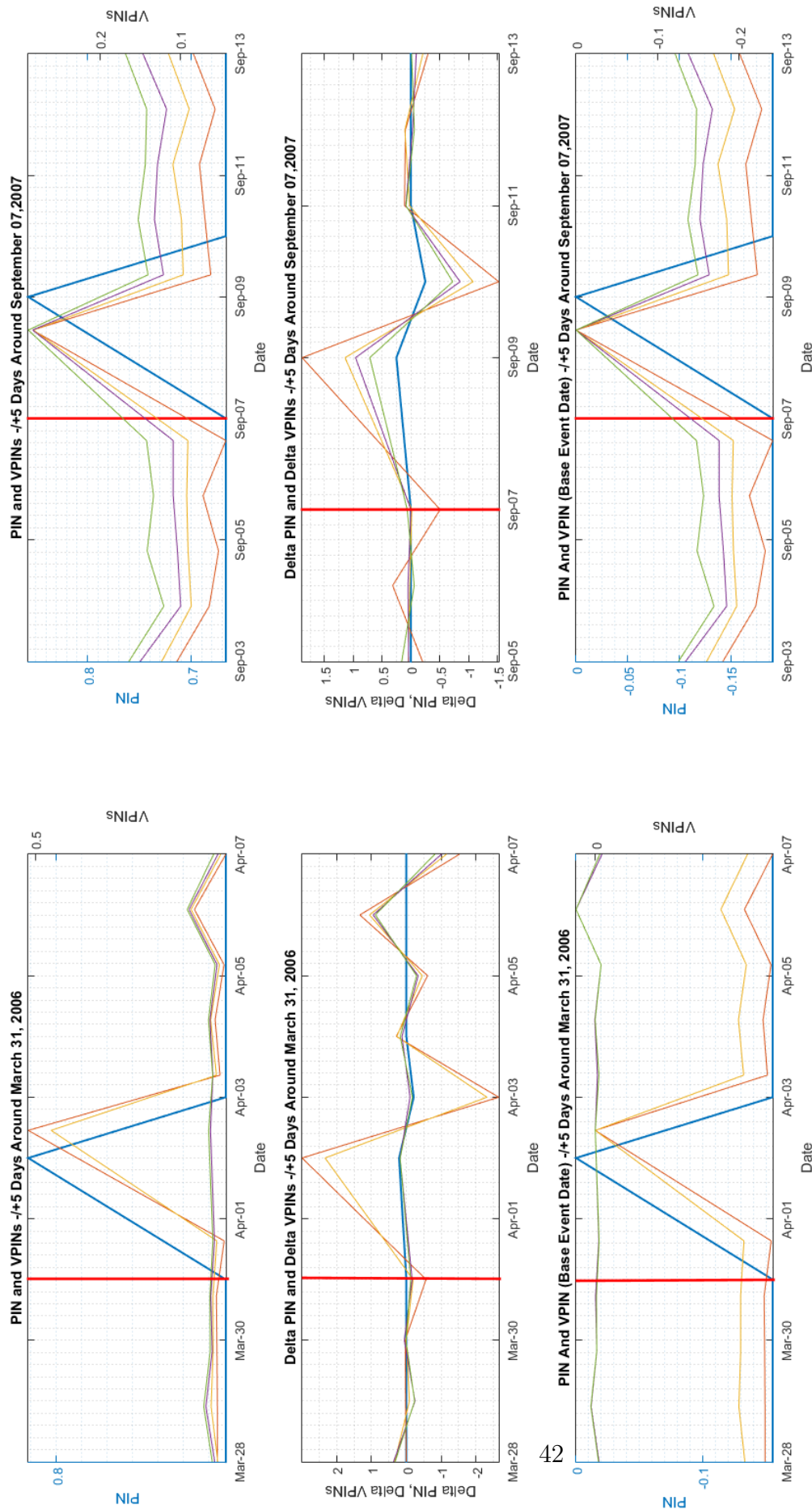


Figure 7: The PIN and VPIN Around Identified LIBOR Manipulation.

Note A: The left panel of figure 7 presents historical PIN on the left axis with a blue line and VPIN on the right axis for the LIBOR manipulation event, between March 28 and April 07, 2006. According to the CFTC document, Barclays' traders attempted to manipulate the US Dollar LIBOR on March 31, 2006, when Barclays' employees, including senior Treasury managers requested the Barclays' LIBOR submitter to submit a higher US Dollar LIBOR rate than the normal rate.

Note B: The right panel of figure 7 presents the PIN on the left axis with a blue line and VPIN on the right axis for LIBOR manipulation event between September 04 to 14, 2007. In this event, Rabobank's US Dollar LIBOR traders asked their submitter on September 07, 2007 to keep a high LIBOR rate for the rest of that week.

VPIN drops to its previous level.

I find PIN and VPIN have a systematic pattern around the LIBOR manipulation events as they increase on the event date, reach a peak the day after, and finally decrease to the previous level. However, it is interesting to note that normally all 40 Eurodollar futures contracts trade within the same time frame. As a result, informed traders may manipulate the ED in different contracts. The limitation of this study is that I cannot identify specific Eurodollar futures contract undertaken by specific banks as this information is not publicly available. Hence, the appropriate way to study PIN and VPIN is to investigate via a term structure and a term structure of variation.

Figure 8 presents the term structure of PIN (Top-Panel) and VPIN (Lower-Panel) for $-/+60$ days around LIBOR manipulation dates. This figure presents the overlay 2,080 plots of PIN and 8,320 plots of VPIN (dotted line) with average PIN and VPIN (thick black line), and also presents 95% confidence interval (black dashed line) into one plot to investigate how the PIN and VPIN react to LIBOR manipulation. The top panel of Figure 8 illustrates that the PIN has a 20% variation from 0.55 to 0.75. The cumulative differences of PIN or cumulative Δ PIN has around a 50% variation of -0.02 to 0.50. However, the average PIN only slightly varies with the highest on the event day from 0.68 to 0.64. The cumulative average Δ PIN also has a small variation between 0.02 and 0.08 with the highest on the event date.

The lower panel of Figure 8 presents the term structure of VPIN and cumulative Δ VPIN with cross sectional average (thick line) across all events, all ED contracts and all types of VPIN (dotted line). It can be seen that the VPIN has higher variation than the PIN. The term structure of VPIN during the event has a minimum value of 0.02 and a maximum of 0.35 or around 35% variation. However, if I focus on the 95% confidence interval, I find VPIN at around 12% variation, mostly moving between 0.10 to 0.20. The average VPIN during LIBOR manipulation varies between 0.135 to 0.16 with the highest VPIN one day

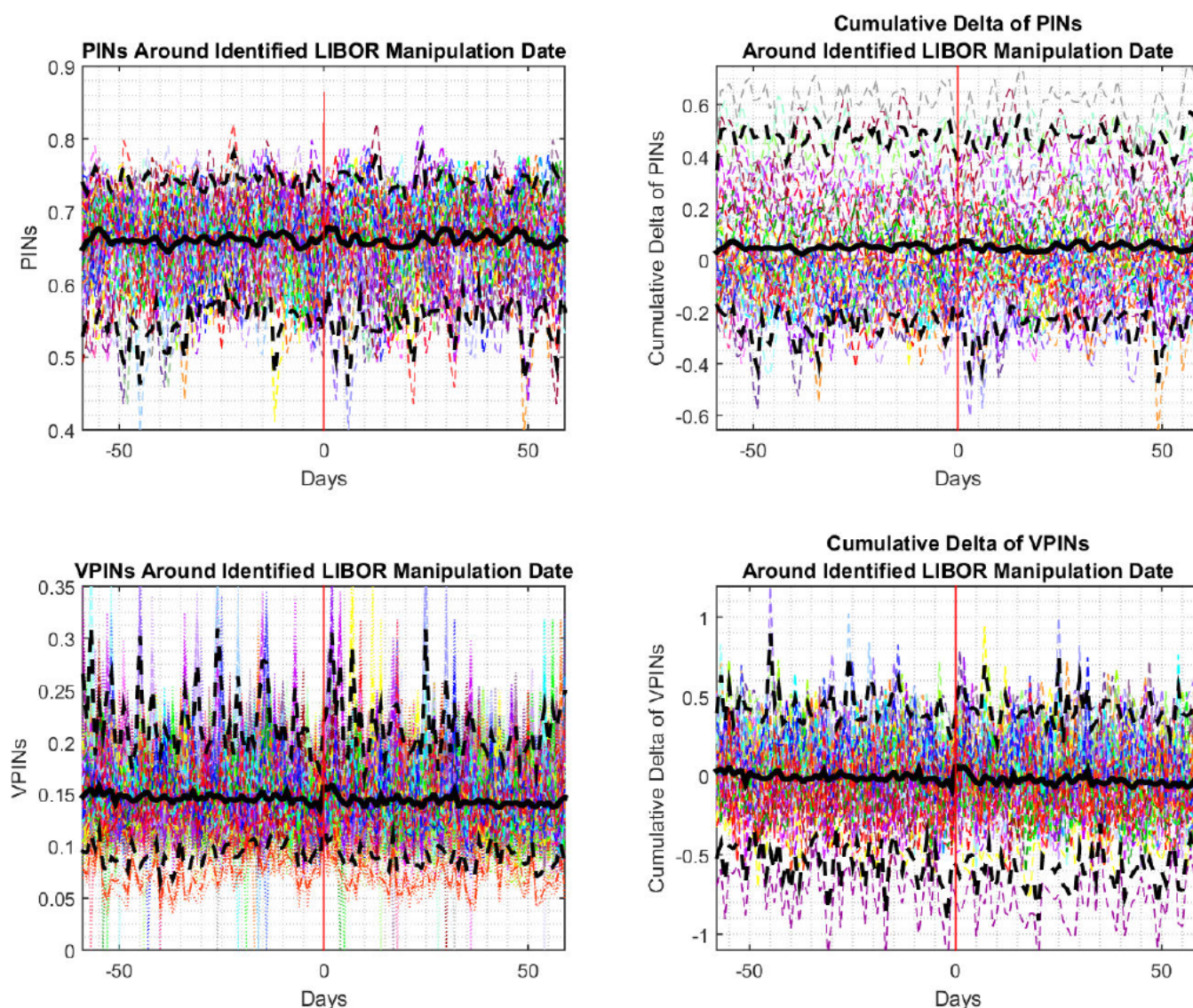


Figure 8: The Term Structure of PIN, VPIN, Delta PIN and Delta VPIN for ± 60 days around identified LIBOR manipulation date.

Note: This figure presents the term structure of PIN and VPIN for ± 60 days around the identified LIBOR manipulation date with the overlay 2,080 plots of PIN and 8,320 VPIN presents on a dotted line, the average PIN and VPIN on a thick black line, and also 95% confident interval on a black dashed line. It can be seen from this figure that the variation of PIN and VPIN around the LIBOR manipulation event in the average daily value is slightly fluctuates.

For the upper panel, the average PIN slightly fluctuates between the lowest at 0.65 and the highest at 0.68. Also, the PIN increases on the event date, then peaks between one or two days after the event date (day zero). Finally, the PIN drops to normal three days after the event.

In the lower panel of figure 8 presents individual VPIN on a dotted line and average VPIN on a thick black line. It can be seen that the average VPIN slightly fluctuates between the lowest at 0.13 and the highest at 0.18. When the PIN is at a peak after the event, the VPIN spiked on the event date then drops between two and three days after the event.

after event at 0.16 and the second highest on event day at 1.58. Finally, the average VPIN drops to normal around three days after the event. One remarkable result is the variation witnessed is lowest from the event date to two days later. This shows that during LIBOR manipulation all types of VPIN have a similar pattern, as they rapidly increase on the event date, remain high for two days then drop to normal.

The right subplot of the lower panel of Figure 8 presents the cumulative Δ VPIN. In this plot, the cumulative Δ VPIN has variation between -0.5 to 0.5 with average value at -0.0194. Also, the highest cross sectional average VPIN on the event day, at 0.07. This increases around 10% compared with the average cumulative Δ VPIN. This result confirms that VPIN has more sensitivity to toxicity events, as the average VPIN and the average cumulative Δ VPIN on the event date are higher than their average. In addition the Δ VPIN is constantly high for at least two days after the event, then drops to normal.

Overall, Figure 8 demonstrates that PIN and VPIN are able to detect toxicity events in the Eurodollar Futures market as they spike around the manipulation date. However, Table 8 shows that the cross sectional average PIN and VPIN have no statistically significant differences around these events, both pre and post 2008. In the pre-2008 period, the highest PIN appears for 60 days after the event and the lowest for 60 days before at 0.7069 and 0.6938 respectively. The highest VPIN in this period appears on VPIN50 for 30 days before the event and the lowest is VPIN20 for 60 days before the event at 0.1713 and 0.0487 respectively. For the post 2008 period, the highest PIN appears for 60 days before and the lowest for 60 days after the event at 0.6752 and 0.6663, respectively. The highest VPIN is VPIN200 for 30 days and, on the contrary, is VPIN20 for 60 days at 0.1773 and 0.0722 respectively. This shows how PIN and VPIN can act partially as an early warning signal for toxic events.

Table 9 below illustrates PIN and VPIN based on event dates both pre and post 2008. In the case of pre 2008, the first column shows that the PIN after day zero is higher than the

Table 8: PIN, VPIN20, VPIN50, VPIN100, and VPIN200 -/+ 60 days around LIBOR manipulation events date.

Days	Informed trading around LIBOR manipulation events date											
	PIN-Pre2008			VPIN - Pre 2008			PIN-Post2008			VPIN - Post 2008		
	20	50	100	200	50	100	200	50	100	200	50	100
-60	0.6938 (0.0196)	0.0487 (0.0584)	0.0600 (0.0590)	0.0652 (0.0505)	0.0722 (0.0400)	0.6752 (0.0089)	0.0490 (0.0350)	0.0656 (0.0356)	0.0815 (0.0360)	0.0917 (0.0329)		
-30	0.6946 (0.0237)	0.1625 (0.0630)	0.1713 (0.0633)	0.1188 (0.0563)	0.0810 (0.0413)	0.6724 (0.0089)	0.1400 (0.0297)	0.1691 (0.0305)	0.1773 (0.0285)	0.1896 (0.0290)		
-20	0.6957 (0.0260)	0.0868 (0.0662)	0.0972 (0.0668)	0.1042 (0.0566)	0.1193 (0.0445)	0.6717 (0.0088)	0.0925 (0.0332)	0.1132 (0.0337)	0.1192 (0.0328)	0.1187 (0.0325)		
-10	0.6967 (0.0287)	0.0493 (0.0623)	0.0886 (0.0627)	0.0648 (0.0537)	0.0706 (0.0412)	0.6731 (0.0081)	0.0741 (0.0338)	0.0993 (0.0344)	0.1121 (0.0336)	0.1259 (0.0330)		
0	0.6968 (0.0325)	0.0500 (0.0510)	0.0573 (0.0627)	0.0673 (0.0499)	0.0724 (0.0375)	0.6732 (0.0117)	0.0578 (0.0332)	0.0786 (0.0327)	0.0991 (0.0326)	0.1174 (0.0311)		
10	0.6995 (0.0324)	0.1396 (0.0364)	0.0602 (0.0500)	0.0814 (0.0302)	0.0884 (0.0303)	0.6738 (0.0053)	0.1241 (0.0290)	0.1447 (0.0266)	0.1635 (0.0259)	0.1781 (0.0209)		
20	0.7020 (0.0318)	0.0570 (0.0447)	0.1519 (0.0376)	0.0712 (0.0432)	0.0780 (0.0451)	0.6686 (0.0053)	0.0684 (0.0290)	0.0947 (0.0266)	0.1126 (0.0250)	0.1314 (0.0230)		
30	0.7043 (0.0322)	0.0549 (0.0492)	0.0650 (0.0457)	0.6590 (0.0435)	0.0701 (0.0397)	0.6671 (0.0043)	0.0700 (0.0281)	0.0997 (0.0271)	0.1271 (0.0252)	0.1483 (0.0228)		
60	0.7069 (0.0350)	0.0911 (0.0425)	0.0593 (0.0412)	0.1263 (0.0374)	0.1343 (0.0338)	0.6663 (0.0017)	0.0995 (0.0316)	0.1128 (0.0297)	0.1288 (0.0288)	0.1389 (0.0284)		

Note: This table compares the average PIN, VPIN20, VPIN50, VPIN100, and VPIN200 including its standard deviation for -/+60, 30, 20 and 10 days around the identified LIBOR manipulation event according to the CFTC and FSA documents from 1996 to the end of 2007, and from 2008 to the end of 2013. It can be seen from this table, that there is a small scale of variation before and after events date for both PIN and VPIN. However, the variation for both parameters for the first sub period(Pre-2008) is higher than post-2008. Despite the fact that there is not a statistically significant difference for PIN and VPIN, both , on the event date, are high compared to ten days before the event.

previous period as the PIN is monotonically increasing from -0.0030 for sixty days before the event and reaches a peak 60 days after, when the PIN is 0.0101 higher than the PIN on event day. The second four columns present four different types of VPIN in the same period as PIN. Unlike the PIN, VPIN is not monotonically increasing. However, all four types of VPIN increase after day zero then drop from three to ten days after the event. The highest VPIN based on the event date is on VPIN20 for ten days after the event, at 0.0896 higher than event day.

Post 2008, the PIN and VPIN have a similar pattern as they rapidly increase on the event date and are constantly high for two to three days, then drop back to normal. The highest PIN appears for ten days after the event at 0.0006, higher than the level on the event date. However, the PIN in this period is lower than pre 2008. The highest VPIN based on the event date appears for 30 days to maturity at 0.0822, 0.0905, 0.0782 and 0.0722, higher than the event date, for VPIN20 follow by VPIN50, VPIN100 and VPIN200 respectively. Both Table 8 and 9 demonstrate that the PIN is weaker than VPIN as a signal of market manipulation post 2008. For example, the average VPIN ten days after the event is mostly 0.06 or 6% higher than the VPIN on the event date. This is compared to 0.0027 or 0.27% for the PIN. However, overall there is no statistically significant difference for PIN and VPIN based on the event date around these events, for either pre or post 2008.

Despite the PIN and VPIN not strongly capturing the market manipulation for -/+ 60 days from the event date, focus on the variation of PIN and VPIN for -/+ 10 days, shows evidence of market manipulation, as this toxicity event is short lived. In general, ten days after the event PIN and VPIN are higher than their level on the event date. Especially in the post 2008, all types of VPIN perform well compared to the PIN. Focus on variation of PIN and VPIN based on the event date, gives us more information as PIN and VPIN for 10 days after the event are always higher than the level on the event date.

The results from table 8 and 9 suggest conclusively, the PIN and VPIN are sensitive to

Table 9: PIN, VPIN20, VPIN50, VPIN100, and VPIN200 based on events date -/+ 60 days around identified LIBOR manipulation events.

Day	Informed trading around LIBOR manipulation events date(based event date)									
	PIN-Pre2008		VPIN - Pre 2008		PIN-Post2008		VPIN - Post 2008			
	20	50	100	200	20	50	100	200		
-60	-0.0030 (0.0196)	-0.0013 (0.0584)	0.0027 (0.0590)	-0.0021 (0.0505)	-0.0002 (0.0400)	0.0020 (0.0089)	-0.0088 (0.0350)	-0.0130 (0.0356)	-0.0176 (0.0360)	-0.0257 (0.0329)
-30	-0.0022 (0.0237)	0.1125 (0.0630)	0.1140 (0.0633)	0.0515 (0.0563)	0.0086 (0.0413)	-0.0008 (0.0089)	0.0822 (0.0297)	0.0905 (0.0305)	0.0782 (0.0285)	0.0722 (0.0290)
-20	-0.0011 (0.0260)	0.0368 (0.0662)	0.0399 (0.0668)	0.0369 (0.0566)	0.0469 (0.0445)	-0.0015 (0.0088)	0.0347 (0.0332)	0.0346 (0.0337)	0.0201 (0.0328)	0.0013 (0.0325)
-10	-0.0001 (0.0287)	-0.0007 (0.0623)	0.0313 (0.0627)	-0.0025 (0.0537)	-0.0018 (0.0412)	-0.0002 (0.0081)	0.0163 (0.0338)	0.0207 (0.0344)	0.0130 (0.0336)	0.0085 (0.0330)
0	0 (0.0325)	0 (0.0510)	0 (0.0627)	0 (0.0499)	0 (0.0375)	0 (0.0117)	0 (0.0332)	0 (0.0327)	0 (0.0326)	0 (0.0311)
10	0.0027 (0.0324)	0.0896 (0.0364)	0.0029 (0.0500)	0.0141 (0.0302)	0.0160 (0.0303)	0.0006 (0.0053)	0.0663 (0.0290)	0.0661 (0.0266)	0.0644 (0.0259)	0.0607 (0.0209)
20	0.0052 (0.0318)	0.0070 (0.0447)	0.0946 (0.0376)	0.0039 (0.0432)	0.0056 (0.0451)	-0.0046 (0.0053)	0.0106 (0.0290)	0.0161 (0.0266)	0.0135 (0.0250)	0.0140 (0.0230)
30	0.0075 (0.0322)	0.0049 (0.0492)	0.0077 (0.0457)	0.5917 (0.0435)	-0.0023 (0.0397)	-0.0061 (0.0043)	0.0122 (0.0281)	0.0211 (0.0271)	0.0280 (0.0252)	0.0309 (0.0228)
60	0.0101 (0.0350)	0.0411 (0.0425)	0.0020 (0.0412)	0.0590 (0.0374)	0.0619 (0.0338)	-0.0069 (0.0017)	0.0417 (0.0316)	0.0342 (0.0297)	0.0297 (0.0288)	0.0215 (0.0284)

Note: This table presents PIN, VPIN20, VPIN50, VPIN100 and VPIN200 based on their value on the event date for -/+ 60, 30, 20, 10 days around the event. The full data is separated into two parts: firstly, 1996 to the end of 2007 and secondly from 2008 to the end of 2013. Both PIN and VPIN based on its value on the event date for all sub-periods is high after the event compared with the previous period. In this case, it can be assumed that the PIN and VPIN somewhat detected some informed trading in the ED futures market on the LIBOR event, as these two parameters are high on the event date, then drops to normal. Finally, the VPIN may have better performance than the PIN, it can be seen in this table that all types of VPIN show a higher positive variation than the PIN on the event date. However, there is not a statistically significantly difference for either based on their value on the event date for the whole period.

toxicity events or informed trading in the futures market, as both increase on the LIBOR manipulation event date. However, these parameters still have limited ability to detect any informed trading as the statistics show no significant difference before and after the event date for both pre and post 2008.

5.6. PIN, VPIN and the Maturity Effect

This section contains an illustration of PIN and VPIN before and after expiration to investigate the relationship between the indicators around the maturity date known as the maturity effect. The most striking result from this figure is an aggressive decline after the spike of PIN and VPIN around the maturity date. Both gradually increase and reach a peak within one week of the last trading day. When the ED starts trading again, they continuously decrease to a lower level, although increase later. However, there is some variation between PIN and VPIN, the VPIN continually decreasing over a longer period than the PIN, and finally slightly increasing around thirty days after the event when the ED begins trading again.

Figure 9 presents the performance of PIN and VPIN around the maturity date. In this event, I use the EDH0 contract which has expiration on March 17, 2010 as an example. This figure is separated into four subplots, first, PIN and VPIN: second, delta PIN and VPIN: third, PIN and VPIN(based on event date) ± 7 days from maturity date: finally PIN and VPIN(based on event date) ± 30 days from maturity date.

On the top panel of the left column, the PIN is seen on the left axis and VPIN on the right for ± 7 days around expiration. There is similar variation between both as the two measurements increase before the maturity date then decrease during the five trading days before expiration. Finally, they bounce back. The highest PIN in this contract appears on March 15, 2010 which is two days to maturity at a value of 0.82 then rapidly decreases to 0.70 on the last trading day. After EDH0 starts trading on the following day, the PIN continuously decreases to around 0.50 and stays constantly low, then finally bounces back

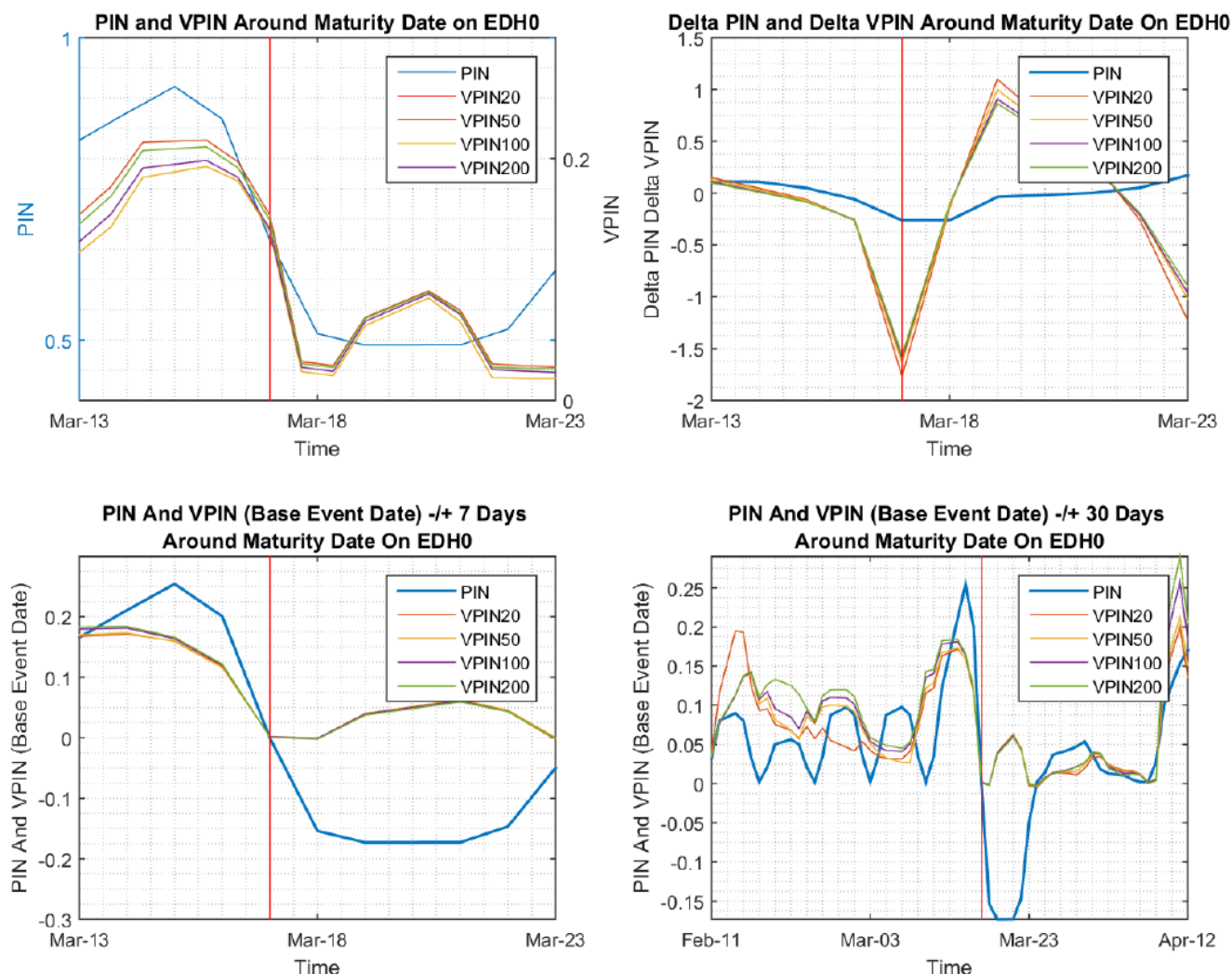


Figure 9: The historical PIN and VPIN of EDH0 with the expiry date.

Note: This figure presents an example of PIN, VPIN, Δ PIN, and Δ VPIN for the EDH0 futures contract. It can be seen from the top left panel that the PIN and the VPIN increase before the maturity date, then rapidly drop around two to three days before the last trading day. Finally, the PIN and VPIN bounce back to normal after the expiration when EDH0 starts trading again on the following day. The top right panel presents Δ PIN and Δ VPIN. This sub-figure illustrates that the differences of VPIN are higher than that of PIN. A high level of Δ VPIN assumes that the VPIN has a higher degree of sensitivity to this toxicity event than the PIN. The lower panel presents PIN and VPIN based on their value on the event date for ± 7 and 30 days from the maturity date. Both show a strong variation around this event compared to its value on the LIBOR event. Clearly, there is a significant pattern as the PIN and VPIN increase, then reach a peak within the last week of the trading period, and dropping to normal the following day.

five day later. Moreover, VPIN is somewhat similar to PIN, the highest VPIN appears around two to three days before expiry at 0.25 then decreases to 0.125 on the last trading day with the lowest at 0.025 between one to two days after the ED starts trading again, and finally bounces back.

The top panel of the right column presents the differences between PIN and VPIN or Δ PIN and Δ VPIN for $-/+7$ days around the maturity date. Despite, Δ PIN and Δ VPIN having a similar pattern, the Δ VPIN has better performance as it has higher variation than PIN. This subplot shows the latter has a small drop of around 10% as the Δ PIN slides from 0 to -0.02 then bounces back to normal on the following day. However, the VPIN drops around one and a half times within seven days compared to the previous period, as the Δ VPIN falls from 0 to -1.5, then rebounds to a level around two times higher than its lowest point. Additionally, the lower panel of Figure 9 presents PIN and VPIN based on the event date for $-/+7$ and $-/+30$ days around the event date. Similar to the upper panel, there are similar patterns for both PIN and VPIN. The VPIN performs better than PIN as a signal of market manipulation, however, the PIN shows a higher variation for the manipulation event.

To gain further insight into the effect of maturity date on PIN and VPIN, both have been investigated for a longer period. Figure 10 illustrates cross sectional averages across all 40 ED contracts on PIN and VPIN for $-/+30$ and $-/+60$ days around the expiry date. Overall, there is a similar pattern to Figure 9. The PIN and the VPIN reach a peak within the last week of the trading period, then drop. Finally, PIN and VPIN bounce back to normal after the last trading day when the ED starts trading again. It can be seen from the top left panel of Figure 10 that from ten days to maturity, the average PIN increases from 0.67 to 0.725 then drops to 0.70 around two days after the last trading day. Also, similar to the PIN, the VPIN increases from 0.07 to around 0.10 three days before the maturity date, then drops to 0.06 two days after. Finally, the VPIN slightly increases to a normal level.

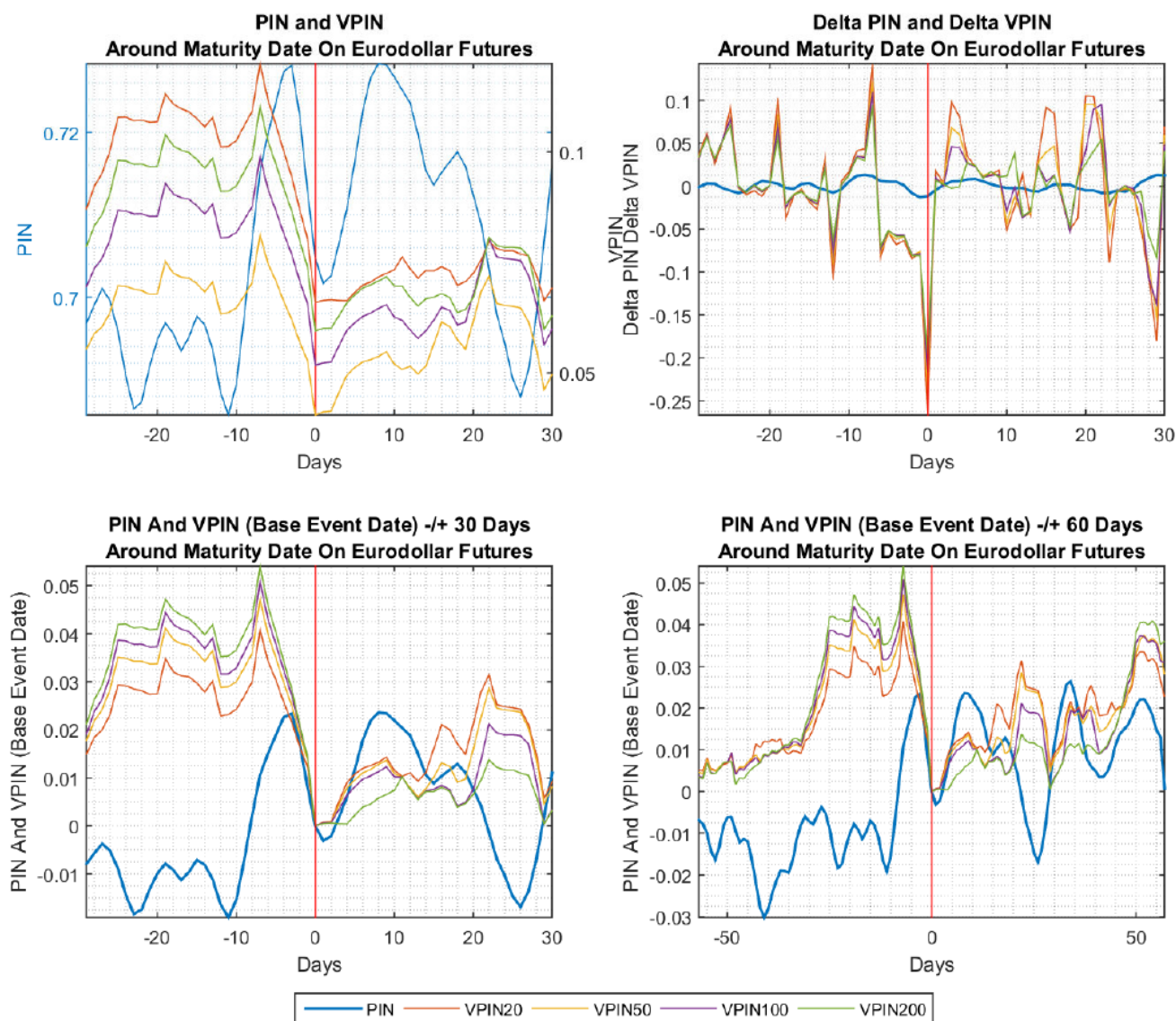


Figure 10: The cross-sectional PIN, VPIN, Δ PIN and Δ VPIN across all 40 ED futures contracts with the expiry date.

Note: This figure presents cross sectional average PIN, VPIN, Δ PIN and Δ VPIN across all 40 ED futures contracts around the expiry date. The red solid vertical line indicates the last trading day on the ED futures contracts. The top left panel presents PIN and four different types of VPIN for ± 30 days around maturity date. During the last week of the trading period, the PIN gradually increases then drops around two to three days until the last trading day. Finally, the PIN and VPIN bounce back. There is a significant pattern between these two parameters. However, the VPIN shows a slightly different pattern from the PIN, as it gradually increases for 30 days before the last trading day, reaching a peak within the last week before the maturity date. The PIN rapidly increases from eleven days to maturity then peaks around two to three days to maturity. Finally, both bounce back after the last trading day when the ED commences trading again. The top right panel presents the differences of PIN and the differences of VPIN (Δ PIN and Δ VPIN). This figure shows a higher degree of difference on the VPIN(Δ VPIN) than the Δ PIN. In this case, I can assume that the VPIN has more sensitivity than the PIN on this event. The lower panel presents PIN and VPIN based on the event date for ± 30 and ± 60 days from maturity date. These two plots show a similar result as the top panel. There is a systematic trend between PIN and VPIN based on their value on the event date and the latter is more sensitive to this event than the PIN.

Top right panel of Figure 10 presents the differences of PIN and VPIN or Δ PIN and Δ VPIN for $-/+30$ days around the maturity date. In this period the Δ VPIN has higher variation than Δ PIN. The Δ PIN fluctuates between -0.01 and 0.01, however, the Δ VPIN fluctuates around 40% between -0.25 and 0.15. The highest Δ VPIN appears on the seven days before the last trading day at 0.15, then gradually drops to the bottom on the expiry date at -0.25. Finally, the VPIN bounces back to normal. Additionally, from seven days to maturity Δ VPIN increases around 25%, from -0.1 to 0.15. Whilst these two parameters have slightly different variation before the maturity date, they have a similar pattern. The Δ PIN and Δ VPIN reach a peak within the last week of the trading period, then rapidly decrease, and finally bounce back after the last trading day when the ED futures start trading the following day.

The lower panel of Figure 10 presents PIN and VPIN based on the PIN and the VIN on the event date for $+/- 30$ and $+/-60$ days around maturity date. These sub-figures show a similar pattern PIN and VPIN based on the event date and normal PIN and VPIN. Additionally, there is a similar trend between both types. The VPIN is constantly high sixty days before expiry, then reaches a peak at around seven days before expiration, as the VPIN based on the event date shows a positive result before the last trading day. However, later between two to three days to the expiration, PIN and VPIN gradually decrease continuously dropping after the last trading day. Finally, there is a move back to normality. However, the PIN based on its value on the event date shows a somewhat negative value from 60 to 5 days to maturity. However, they reach a peak within one week before the maturity date, then rapidly drop.

The following section presents the result of the term structure of PIN and VPIN on the maturity effect as presented for the LIBOR manipulation events. Figure 11 presents the term structure of PIN and VPIN, also cumulative Δ PIN and cumulative Δ VPIN for the maturity effect. The figure presents individual PIN, VPIN20, VPIN50, VPIN100, and

VPIN200 as a dotted line and average PIN and VPIN as a thick line. It also presents 95% range as a 95% confident interval as a thick dotted line. Overall, I present 40 PIN and 200 VPIN for the term structure.

The top panels of Figure 11 present cross-sectionals of PIN and cumulative Δ PIN. Overall, there is a small variation on average PIN around the maturity date with a small spike about three days to expiration. This spike has a variation of 2% higher than the average. Despite there being a small scale of variation on average PIN before the maturity date, the individual PIN has a higher variation after the last trading day. The major trend of PIN (dotted lines) is constantly moving around the identical PIN at about 0.67 (thick line) with a small spike and there is the widest variation of PIN at three days to expiration. The lowest PIN is 0.2 and the highest is 0.995, which is nearly 80% variation.

However, after three days to expiration the variation of PIN drops to 50%, with the highest recording at 0.97 and the lowest around 0.30. The variation is wider at the beginning than the previous trading period, this may be consistent with the term structure of PIN (figure 4) as it has a high variation from the first trading day; then the variation becomes narrow when the ED futures are nearer to maturity.

The lower panel of Figure 11 illustrates VPIN and cumulative of Δ VPIN for -/+60 days from the maturity date across all 40 ED futures contracts. This panel presents not only forty individual VPIN and forty cumulative Δ VPIN from all ED contracts, but each contract also calculates four different types of VPIN and cumulative Δ VPIN presented on a dotted line with the average value in a thick line, and 95% confident interval in a thick dotted line. It can be seen from this figure that there is quite naturally the same pattern of average of VPIN with the PIN. The average VPIN gradually increases from 60 days from the last trading days and reaches a peak at around two days to maturity, then rapidly drops. Finally, the VPIN increases to a normal level after the expiration when the ED start trading again. Also, there is a lower scale of variation on VPIN than PIN, which is notably smaller and

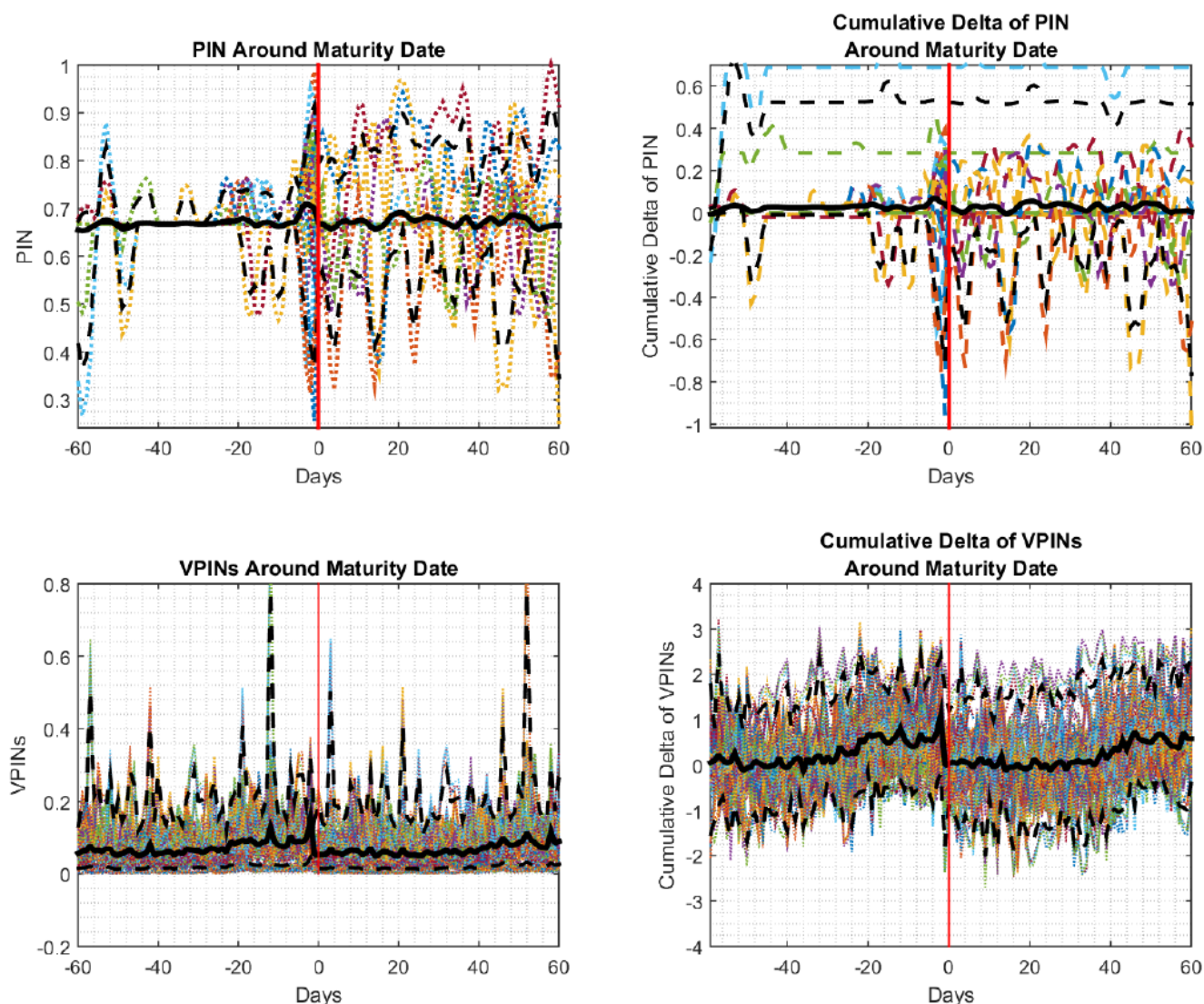


Figure 11: The variation of PIN and VPIN -/+60 days around maturity date

Note: This figure presents the term structure of PIN, VPIN, cumulative of Δ PIN and cumulative of Δ VPIN with their average value for -/+60 days around maturity date. Overall, there is a systematic pattern between the average PIN and VPIN around this event, as these parameters show a spike on two days to maturity then rapidly drop on the last trading day. Finally, these two parameters bounce back after the last trading day when the ED futures start trading again. However, the lower panel shows that all types of VPIN have a higher scale of variation than the PIN for both before and after the maturity date.

more systematic than the PIN. The lowest VPIN is near to zero and the highest is about 0.8, with the average upper bound at about 0.2.

In comparison, the average PIN has a small variation around 0.67 and the average cumulative Δ PIN is near to zero. However, the average VPIN gradually increases from 0.05 to just above 0.20 and the average cumulative Δ VPIN increases from just around zero to one on two days to the maturity date. Finally, VPIN and cumulative Δ VPIN plunges to nearly zero on the last trading day. The average VPIN and the average cumulative Δ VPIN reaches a peak on two days to expiration which increases by around twofold, compared to the previous period. the average VPIN has a higher variation than the PIN and there is a systematic trend between different VPINs. One final concluding remark is that the term structure of Δ VPIN tends to be a 'sine curve'.

After studying PIN and VPIN via the term structure around the maturity date, I then investigate these two parameters from their standard deviation(SD). Table 10 reports cross sectional average PIN and VPIN across all 40 ED contracts, constructed for 60, 30, 20 and 10 days for fixed time windows around the maturity dates with their SD. Then, the result of these two parameters is divided into two sub-periods. The first sub-period is between January 1, 1996 and December 31, 2007 and the second sub-period is between January 1, 2008 until December 31, 2013. Moreover, this time period is separated into two parts because there will be different results for PIN and VPIN between these two periods as there is the break in the relationship of the bank rank of LIBOR quotes after 2008.

Overall, the variation of PIN is lower than VPIN for both periods as its standard deviation is smaller than the VPIN. However, there is a similar pattern between these two parameters. They both increase before the last trading day, then rapidly drop. Finally, they bounce back after the last trading day when the ED futures starts trading again. The lowest PIN appears on the last trading day (Day 0) at 0.6695 and 0.6552 for pre 2008 and post 2008 samples respectively. The highest appears for 20 and 60 days after the maturity

date for pre-2008 and post-2008 respectively. For the VPIN, the lowest appears on VPIN20 for both pre and post 2008, at 0.0471 and 0.0309 respectively. The highest VPIN mostly appears for thirty days to maturity for all different types of VPIN, with the highest of 0.1639 on VPIN200 for pre 2008. This table indicates that there is a similar trend for PIN and VPIN for both periods, however, their variation and standard deviation (SD) are different.

Pre 2008, the SD on PIN and VPIN are slightly different. The SD on PIN shows that the PIN is not statistically significantly different, as the SD varies between 0.02 to 0.05. However, the SD on VPIN during this period is slightly higher than PIN. The highest SD for 10 days after the maturity date is 0.088, 0.089, 0.095 and 0.099 for VPIN20, VPIN50, VPIN100 and VPIN200 respectively. Post 2008, the highest SD for PIN is 0.055 for 60 days after the maturity date. However, the highest SD for VPIN is for 30 days to maturity at 0.108, 0.0833 and 0.085 for VPIN20, VPIN100 and VPIN200 respectively. The highest SD for VPIN20 is 0.127 for ten days to maturity which is the highest SD for all types of VPIN. Finally, there are two remarkable results from this table; first the maturity effect tends to have an impact on informed trading as the PIN and VPIN have shown a similar pattern. They slightly increase for 60 days to maturity, then fall to the lowest PIN and VPIN on the last trading day. Next, the smallest VPIN on this event for both the pre 2008 and post 2008 period, is VPIN20 followed by VPIN50, VPIN100 and VPIN200.

Next, I analyze PIN and VPIN based on their value on the event date for $-/+60$ days around the maturity which is presented in Table 11. The results indicate that the PIN and VPIN around the maturity date are higher than the those on regular trading days. Using the same format analysis as Table 10, I find the same result. First, the VPIN has a greater magnitude of variation than the PIN for both pre and post 2008. Secondly, the PIN and VPIN gradually increase for 60 days to maturity then peak within five trading days. After three to two days to maturity, they rapidly drop and finally bounce back. On pre-2008, the highest PIN is for ten and twenty days after the maturity date at 0.0283 or 2.83%,

Table 10: PIN, VPIN20, VPIN50, VPIN100, and VPIN200 -/+ 60 days around maturity date.

Days	PIN -Pre 2008	Informed trading around maturity date									
		VPIN - Pre 2008					PIN - Post 2008				
		20	50	100	200		20	50	100	200	
-60	0.6926 (0.0223)	0.1104 (0.0851)	0.1238 (0.0785)	0.1335 (0.0734)	0.1423 (0.0721)		0.6933 (0.0324)	0.0753 (0.0924)	0.0797 (0.0858)	0.0809 (0.0764)	0.0841 (0.0754)
-30	0.6933 (0.0278)	0.1207 (0.0881)	0.1396 (0.0817)	0.1488 (0.0714)	0.1639 (0.0775)		0.6975 (0.0400)	0.0911 (0.1088)	0.0937 (0.0953)	0.0968 (0.0833)	0.1027 (0.0850)
-20	0.6926 (0.0313)	0.1208 (0.0903)	0.1368 (0.0836)	0.1459 (0.0752)	0.1564 (0.0751)		0.6999 (0.0433)	0.0833 (0.1001)	0.0859 (0.0875)	0.0873 (0.0758)	0.0924 (0.0774)
-10	0.6930 (0.0354)	0.1126 (0.0853)	0.1285 (0.0796)	0.1375 (0.0721)	0.1466 (0.0720)		0.7006 (0.0459)	0.0749 (0.0920)	0.0783 (0.1276)	0.0793 (0.0699)	0.0839 (0.0715)
0	0.6695 (0.0368)	0.0471 (0.0734)	0.0591 (0.0682)	0.0724 (0.0662)	0.0827 (0.0639)		0.6552 (0.4656)	0.0309 (0.0767)	0.0384 (0.0726)	0.0417 (0.0626)	0.0444 (0.0616)
10	0.6978 (0.0420)	0.1012 (0.0888)	0.1176 (0.0892)	0.1323 (0.0965)	0.1397 (0.0998)		0.7066 (0.0448)	0.0497 (0.0641)	0.0532 (0.0586)	0.0531 (0.0484)	0.0576 (0.0568)
20	0.6979 (0.0466)	0.0981 (0.0727)	0.1105 (0.0740)	0.1222 (0.0764)	0.1279 (0.0750)		0.7097 (0.0477)	0.0427 (0.0469)	0.0473 (0.0341)	0.0482 (0.0359)	0.0507 (0.0413)
30	0.6970 (0.0488)	0.0931 (0.0665)	0.1049 (0.0666)	0.1170 (0.0700)	0.1223 (0.0658)		0.7116 (0.0501)	0.0501 (0.0522)	0.0556 (0.0517)	0.0563 (0.0471)	0.0572 (0.0489)
60	0.6891 (0.0530)	0.0981 (0.0593)	0.1106 (0.0558)	0.1210 (0.0578)	0.1262 (0.0537)		0.7157 (0.0550)	0.0518 (0.0546)	0.0580 (0.0547)	0.0574 (0.0417)	0.0591 (0.0401)

Note: This table compares average PIN, VPIN20, VPIN50, and VPIN100, with their standard deviation for -/+60, 30, 20 and 10 days from maturity date. The time period is separated into two periods: the first sub-period is from 1996 to the end of 2007 and the second from 2008 to the end of 2013. Overall, there is small scale variation for both PIN and all types of VPIN before and after the event dates. However, their standard deviation pre-2008 are higher than post-2008. Additionally, VPIN has a higher standard deviation than PIN. It can be assumed that VPIN is more sensitive to the maturity effect than PIN, as it increases before the last trading day then drops dramatically by over 50% on the last day. Finally, there is a similar pattern between these two parameters. However, neither are statistically significantly different during the LIBOR manipulation event, as their standard deviation varies between 0.02 to 0.09.

Table 11: The PIN, VPIN20, VPIN50, VPIN100, and VPIN200 based events date -/+ 60days around maturity date.

Days	PIN -Pre 2008	Informed trading around maturity date									
		VPIN - Pre 2008			PIN - Post 2008			VPIN - Post 2008			
		20	50	100	200			20	50	100	200
-60	0.0231 (0.0223)	0.0633 (0.0851)	0.0647 (0.0785)	0.0611 (0.0734)	0.0596 (0.0721)	0.0381 (0.0324)		0.0444 (0.0924)	0.0413 (0.0858)	0.0392 (0.0764)	0.0397 (0.0754)
-30	0.0238 (0.0278)	0.0736 (0.0881)	0.0805 (0.0817)	0.0764 (0.0714)	0.0812 (0.0775)	0.0422 (0.0400)		0.0602 (0.1088)	0.0553 (0.0953)	0.0551 (0.0833)	0.0583 (0.0850)
-20	0.0231 (0.0313)	0.0737 (0.0903)	0.0777 (0.0836)	0.0735 (0.0752)	0.0737 (0.0751)	0.0447 (0.0433)		0.0524 (0.1001)	0.0475 (0.0875)	0.0456 (0.0758)	0.0480 (0.0774)
-10	0.0235 (0.0354)	0.0655 (0.0853)	0.0694 (0.0796)	0.0651 (0.0721)	0.0639 (0.0720)	0.0454 (0.0459)		0.0440 (0.0920)	0.0399 (0.1276)	0.0376 (0.0699)	0.0395 (0.0715)
0	0	0	0	0	0	0		0	0	0	0
10	0.0283 (0.0420)	0.0541 (0.0888)	0.0585 (0.0892)	0.0599 (0.0965)	0.0570 (0.0998)	0.0514 (0.0448)		0.0188 (0.0641)	0.0148 (0.0586)	0.0114 (0.0484)	0.0132 (0.0568)
20	0.0283 (0.0466)	0.0510 (0.0727)	0.0514 (0.0740)	0.0498 (0.0764)	0.0452 (0.0750)	0.0545 (0.0477)		0.0118 (0.0469)	0.0089 (0.0341)	0.0065 (0.0359)	0.0063 (0.0413)
30	0.0275 (0.0488)	0.0460 (0.0665)	0.0458 (0.0666)	0.0446 (0.0700)	0.0396 (0.0658)	0.0564 (0.0501)		0.0192 (0.0522)	0.0172 (0.0517)	0.0146 (0.0471)	0.0128 (0.0489)
60	0.0196 (0.0530)	0.0510 (0.0593)	0.0515 (0.0558)	0.0486 (0.0578)	0.0435 (0.0537)	0.0605 (0.0550)		0.0209 (0.0546)	0.0196 (0.0547)	0.0157 (0.0417)	0.0147 (0.0401)

Note: This table presents a comparison of average PIN, VPIN20, VPIN50, VPIN100, and VPIN200 based on their value on the event date, and also standard deviation (SD) for -/+60, 30, 20 and 10 days from the maturity date. The time period is separated into two periods: the first sub period from 1996 to the end of 2007 and the second is from 2008 to the end of 2013. Overall, there is a similar result to Table 10 - a small scale variation before and after the event dates for both PIN and VPIN. Despite the variation of VPIN being higher for the first pre-2008 period, the variation of PIN for post-2008 is higher than pre-2008. Nevertheless, there is a systematic trend between these two parameters as the PIN and all types of VPIN increase before the last trading day then drop dramatically more than 50% on the last day. Finally, they increase to the normal level.

higher than the PIN on the maturity date. Moreover, the highest VPIN based on the VPIN on the event date appears for thirty and twenty days before the maturity date, at 0.0737, 0.0805, 0.0764 and 0.0812 for VPIN20, VPIN50, VPIN 100 and VPIN200 respectively. For the second sub period, the highest PIN based on the PIN on the event date is for sixty days after the expiration at 0.605 or 6% higher than that on the expiry date. Similar to the first period, the highest VPIN appears for thirty and twenty days before the maturity date at 0.0602, 0.0553, 0.0551 and 0.0583 for VPIN20, VPIN50, VPIN 100 and VPIN200 respectively.

Conclusively, I find the VPIN has a higher scale of variation than PIN except after the maturity date for the second sub period. For the first sub period, the average PIN for sixty days before the maturity date is at 0.023 or 2.3% higher than the last trading day. The average VPIN from sixty days to maturity is at 0.07 or 7% higher than the VPIN on the last trading day, which is around three times higher when compared to the PIN. The clear indication is that both have a similar pattern around the maturity event as they increase then reach a peak before it. Finally, they continuously drop after the last trading day when the ED starts trading again, before bouncing back to normal. Also, the results are in the line with the evidence, as described in the previous analysis section, that the VPIN has more predictability for truly toxicity events than the PIN.

6. Conclusion

This paper takes a comprehensive dataset of the Chicago Mercantile Exchange (CME) tape data, which covers every inside quote and trade from 1996 to 2015 for the 40 LIBOR referenced, quarterly dated Eurodollar futures contracts. First, I apply different types of PIN and VPIN metrics over a variety of estimation windows. I have undertaken an empirical microstructure model of Easley and O'Hara (1992) and Easley et al. (1996) for the PIN and Easley et al. (2011) and Easley et al. (2012) for VPIN. I then have constructed a variety of

tests to see if the pattern of PIN and VPIN exhibit structural changes around documented cases of manipulation of the LIBOR reference rate and the maturity event. Finally, I compare this to systematic fluctuations in these measures relative to the futures term structure.

Unsurprisingly, given the scale of the task, therefore, the results are very mixed. Both PIN and VPIN vary systematically and in a statistically significant pattern in respect to the term structure of the futures contracts. PIN varies in a v-shaped pattern, with long (2000 to 3500 days) and short maturity (0 to 500 days) contracts, having significantly higher PIN than intermediate contracts (which are actually the most heavily traded). However, VPIN tends to be a v-shaped pattern from 900 days to maturity to the last trading day. Similar to the PIN, the VPIN on long and short maturity contracts, has significantly higher PIN than intermediate contracts. The former is substantially lower than the latter over the entire range of calculation measures. However, when I move to documented cases of market manipulation in the LIBOR reference rate, the results are ambiguous. There are definitive examples when the PIN and the VPIN shift systematically around a relevant, documented case of a LIBOR manipulation. However, when I build cross sectional averages across events, there is no significant evidence of systematic shifts in either the PIN or VPIN metric. It should be noted that whilst I have included every documented case of manipulation directly linked to the relevant reference rates, the list is necessarily incomplete as the regulatory actions have tended to focus on sample charges to the firms involved, rather than documenting every occurrence and its motivation. These findings clearly show that, consistent with Andersen and Bondarenko (2014a) Abad et al. (2015), the PIN and VPIN have less predicting power as an early signal warning on market manipulation.

The investigation extends to the maturity effect. The results are remarkable and show PIN and VPIN have a significant pattern around this event. They reach a peak at three days to maturity, then drop within the last week of the trading period. Finally, both gradually increase to a normal level after the ED starts trading again.

Another remarkable result in this study is that the cumulative of Δ PIN and the cumulative of Δ VPIN show a clear pattern on the event study, as they spike on the toxicity manipulation event. In line with Easley et al. (2012), the results also show that the cumulative value of Δ VPIN performs better than the normal value in providing information about toxicity events. Therefore, I also apply this method to the PIN. The cumulative of Δ PIN and the cumulative of Δ VPIN show a clear pattern on the event study, as they spike on LIBOR manipulation dates, which are higher than the spike on the normal PIN and VPIN.

Finally, despite VPIN having a higher deviation than PIN and their cumulative differences performing better than the normal value to capture these toxicity events, they are not significantly statistically different for both the LIBOR manipulation and the maturity event. The results may be because the PIN and VPIN approach is unsuitable or because it may have been used incorrectly for this type of data. Alternatively, the LIBOR manipulation and expiration may have only a minor effect. However, I leave full investigation of this issue to future work, as more fine detail emerges from the current round of court cases and provides more direct evidence of channels of informed trading.

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